



Socius: a beginner's guide

Statistics and text coding for sociologists, in your browser

Edition 1, September 2026

Web version: <https://hackhead95.github.io/socius/guide/>

Socius: <https://hackhead95.github.io/socius/>

Contents

1	Welcome to Socius	3	7	Your first analyses	18
	Who Socius is for	3		Frequencies: how often does each answer occur?	18
	What it costs	3		Crosstabs with chi-square: are two categories related?	20
	Your data stays on your computer	3		Comparing two groups: the independent-samples t-test	22
	What you need	3		Comparing several groups: one-way ANOVA	23
2	A five-minute tour	4		Correlation: do two numbers move together?	25
	Coming back to the start screen	6		A simple linear regression	26
	Small messages and dialogs	6		Where to go next	27
	The sample survey	7	8	Charts	27
3	Finding anything with Search	7		A bar chart	28
	Open Search	7		A histogram	28
	What you can search for	8		Saving a chart	29
4	Opening your data	9	9	Getting results into your report	29
	Open an SPSS file	9		APA style or SPSS style	30
	Open a CSV or Excel file	9		Copy one table into Word	31
5	Understanding your variables	10		Copy the APA sentence	31
	Rename a variable	11		Export the whole report	31
	Value labels	12	10	Coding open-ended answers	32
	Missing values	12		Start with the worked example	32
	Check many variables at once: Define variable properties	13		Import answers from a survey question	32
	Measurement level: nominal, ordinal or scale	14		The Responses view	33
6	Preparing your data	15		Build a codebook	34
	Recode age into groups	15		Code with the number keys	35
	Reverse-code a question	16		Auto-code with keyword rules	36
	Build a scale and check Cronbach's alpha	16		Turn codes into variables and crosstab them	36
	Select cases	17	11	Coding interviews	38
	Weight cases	17		Import a transcript	38
				Highlight and code a passage	38
				Write memos	40
				Retrieve quotes	40
				Intercoder reliability: do two coders agree?	41

12 Getting help from AI	42	13 Saving and sharing your work	57
Set it up	43	Save a project	57
Before you send anything	48	What Socius keeps in your browser	57
Where to find AI in Socius	48	Save your data for SPSS, Excel or R	58
Explain a result	49	Share with a colleague	59
The Socius assistant	51	14 Troubleshooting and FAQ	59
AI in Text coding	56	Troubleshooting with the error log	61
Reporting AI use	56	15 Giving feedback	62
		Appendix A Quick reference card	64
		Appendix B Glossary	66

Welcome to Socius

Socius is a free research workbench for sociologists. It runs in your web browser. You can open SPSS files, run the statistics you know from SPSS, and code interviews and open-ended survey answers, all in one place.

This guide is for you if you are new to Socius, and also if you are new to statistics software. It takes you step by step from opening a file to a finished results section. Every example uses a practice survey that comes with Socius, so you can follow along on your own computer.

Who Socius is for

- **Students** writing a dissertation or a methods assignment.
- **Researchers** who analyse survey data and interviews.
- **SPSS users** who want the same menus and tables without a licence. The Data View, Variable View, dialogs and output tables look and work much like SPSS.

What it costs

Nothing. Socius is free to use. There is no account, no sign-up and no trial period.

Your data stays on your computer

Socius runs entirely inside your browser. When you open a file, it is read on your computer and is never uploaded anywhere. Socius keeps a copy of your current work in the browser so it is there when you come back, but that copy lives only in this browser on this computer. To keep your work safe, save a project file (see Saving and sharing your work (page 57)). **Help > About Socius** lists everything Socius keeps in the browser (see What Socius keeps in your browser (page 57)).

The one exception is the optional AI help. It is off until you set it up, and it never sends anything until you click a button or ask a question. Each AI feature tells you what it will send, and to whom, before it does (see Getting help from AI (page 42)).

What you need

- A computer with a recent version of **Chrome, Edge, Firefox or Safari**.
- An internet connection to open the Socius page. After that, your data does not travel anywhere.
- A mouse or trackpad and a keyboard. Socius opens on tablets and phones, which is fine for looking at results, but a computer is much easier for real work.

To start, open Socius at <https://hackhead95.github.io/socius/>. You can come back to this guide at any time from **Help > User guide**.

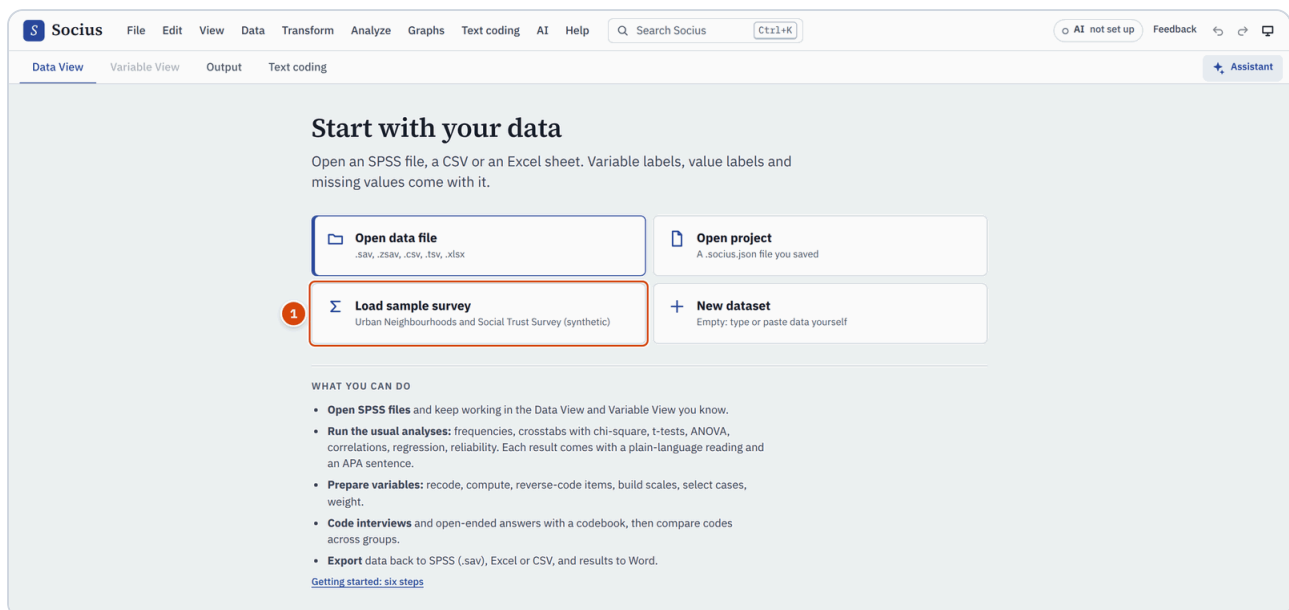
HOW TO READ THIS GUIDE

Menu paths are written like this: **Analyze > Descriptive Statistics > Frequencies**. It means: click **Analyze** in the menu bar, then **Descriptive Statistics**, then **Frequencies**. In the app, items that open a dialog end with "...". Variable names from the practice survey look like this: `life_sat`.

CHAPTER 2

A five-minute tour

When you open Socius for the first time, you see the welcome screen. It offers four ways to start: open your own data file, open a project you saved before, load the sample survey, or type data into an empty dataset.



The welcome screen. Click Load sample survey (1) to follow this guide.

Click **Load sample survey** now. The practice data opens, and you see the main screen.

	resp_id	city	area	interviewer	int_date	gender	age	educ	employ	marital	hh_size	hh_income	
1	1001	Kolkata	Peri-urban	INT01	06-JAN-2025	Man	43	Graduate	Employed...	Married	8	24000	Bor
2	1002	Kolkata	Core city	INT02	06-JAN-2025	Man	19	Postgradu...	Student	Never marr...	4	88500	Mig
3	1003	Kolkata	Peri-urban	INT01	07-JAN-2025	Man	35	Secondary	Unemploy...	Married	6	9000	Bor
4	1004	Kolkata	Peri-urban	INT02	07-JAN-2025	Man	39	Secondary	Employed...	Married	3	10000	Bor
5	1005	Kolkata	Core city	INT03	07-JAN-2025	Woman	44	Secondary	Homemak...	Married	4	31000	Bor
6	1006	Kolkata	Core city	INT02	08-JAN-2025	Man	54	Secondary	Unemploy...	Married	4	14500	Bor
7	1007	Kolkata	Core city	INT03	08-JAN-2025	Man	30	Higher se...	Unemploy...	Never marr...	4	29500	Mig
8	1008	Kolkata	Core city	INT01	09-JAN-2025	Woman	35	Secondary	Employed...	Married	1	8000	Bor
9	1009	Kolkata	Core city	INT02	09-JAN-2025	Woman	26	Higher se...	Employed...	Married	4	9500	Mig
10	1010	Kolkata	Core city	INT03	09-JAN-2025	Man	33	Graduate	Employed...	Married	3	11000	Bor
11	1011	Kolkata	Core city	INT03	10-JAN-2025	Man	29	Graduate	Employed...	Never marr...	2	9000	Mig
12	1012	Kolkata	Core city	INT01	11-JAN-2025	Man	49	Higher se...	Homemak...	Married	9	28500	8
13	1013	Kolkata	Core city	INT02	11-JAN-2025	Woman	44	No formal...	Homemak...	Married	7	8000	Bor
14	1014	Kolkata	Core city	INT02	11-JAN-2025	Man	32	Postgradu...	Employed...	Never marr...	5	43000	Bor
15	1015	Kolkata	Core city	INT01	13-JAN-2025	Woman	50	Higher se...	Homemak...	Married	5	78500	Bor
16	1016	Kolkata	Core city	INT01	13-JAN-2025	Woman	32	Postgradu...	Employed...	Married	6	65000	Bor
17	1017	Kolkata	Core city	INT01	13-JAN-2025	Man	48	Postgradu...	Self-empl...	Married	6	22500	Mig
18	1018	Kolkata	Core city	INT02	13-JAN-2025	Man	26	Graduate	Employed...	Married	4	51500	Bor
19	1019	Kolkata	Core city	INT03	13-JAN-2025	Man	83	Primary	Retired	Married	6	7000	Bor
20	1020	Kolkata	Peri-urban	INT01	14-JAN-2025	Man	58	Primary	Self-empl...	Married	6	12000	Bor
21	1021	Kolkata	Core city	INT02	14-JAN-2025	Woman	42	Graduate	Homemak...	Married	6	49500	Bor
22	1022	Kolkata	Core city	INT02	14-JAN-2025	Other / pr...	82	Higher se...	Homemak...	Widowed	5	Refused	Bor
23	1023	Kolkata	Core city	INT03	14-JAN-2025	Woman	29	Postgradu...	Homemak...	Married	3	92500	Bor
24	1024	Kolkata	Peri-urban	INT03	15-JAN-2025	Man	31	Graduate	Self-empl...	Married	3	21000	Mig
25	1025	Kolkata	Core city	INT01	16-JAN-2025	Woman	69	No formal...	Homemak...	Married	9	8000	Bor
26	1026	Kolkata	Peri-urban	INT01	16-JAN-2025	Man	44	Higher se...	Self-empl...	Married	2	16000	Mig
27	1027	Kolkata	Peri-urban	INT01	16-JAN-2025	Woman	78	Secondary	Retired	Widowed	6	18000	Bor
28	1028	Kolkata	Core city	INT01	16-JAN-2025	Man	40	Graduate	Employed...	Married	4	24500	Mig

The main screen with the sample survey open.

The numbers in the picture show the parts of the screen:

- 1. Home button and menu bar.** The **Socius** logo at the far left is the Home button: it takes you back to the start screen (see below). Everything Socius can do is in the menus next to it: File, Edit, View, Data, Transform, Analyze, Graphs, Text coding, AI and Help. Click a menu to open it; while it is open, point at another menu to open that one instead.
- 2. Search.** Type what you are looking for, such as "chi square" or a variable name, and Socius finds the menu item, variable, result or help page (see Finding anything with Search (page 7)).
- 3. AI, Feedback, Undo, Redo and theme.** The **AI** chip shows whether AI help is set up ("not set up" until you choose an option). **Feedback** opens a short dialog for reporting a problem or suggesting an idea (see Giving feedback (page 62)). Undo and Redo reverse your last change in the tab you are in. The last button switches between light and dark colours.
- 4. Dataset bar.** The name of the open dataset, how many cases (rows) and variables (columns) it has, and small labels that tell you when a weight or a filter is switched on.
- 5. Main tabs.** **Data View** shows the data, one row per respondent. **Variable View** describes the variables. **Output** collects your results. **Text coding** is where you code interviews and open-ended answers.
- 6. Variable list.** All your variables with their labels. Type in the search box to find one quickly. **View** > **Variable list** hides and shows it. In a narrow window (below about 900 pixels wide, such as a tablet), the list is hidden to give the data more room, and **View** > **Variable list** opens it as a drawer over the page.
- 7. Toolbar.** Buttons for the current tab. In Data View you can switch between codes and value labels, add cases or variables, sort, and search.
- 8. Data grid.** The data itself. Each row is a case (usually a respondent) and each column is a variable (usually a question).

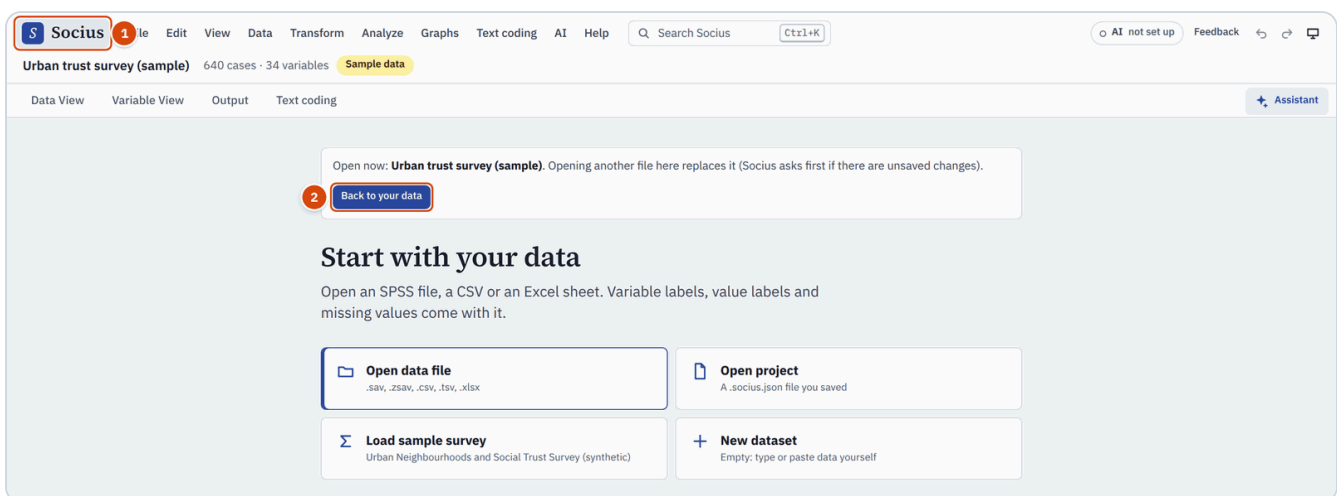
9. **Assistant button.** At the right end of the row of tabs. It opens the Socius assistant, which answers questions about your data and methods once AI help is set up (see The Socius assistant (page 51)). On a phone it shows only its icon.

FOR SPSS USERS

Data View, Variable View and the Output tab work like their SPSS counterparts. The main difference is that output lives in a tab of the same window instead of a separate viewer window.

Coming back to the start screen

Click the **Socius** logo (1) at any time to see the start screen again. From there you can open another data file or project, load the sample survey, or pick a recent project. While something is open, a bar at the top says what it is, for example "Open now: Urban trust survey (sample)". Click **Back to your data** (2), or any tab, to return to your work. (When only output or coding is open, the button says **Back to Output** or **Back to Text coding**.)



The start screen, opened with the Socius logo (1). Back to your data (2) returns to your work.

Opening a file from the start screen replaces the open data, just like **File > Open data file**. If you have unsaved changes, Socius asks first.

Small messages and dialogs

After many actions, a small message appears at the bottom of the screen for a few seconds, for example "Created agegrp by recoding age." Some messages have a button: **Undo** takes the action back, and **Show** takes you to the variable you just made. If the same message appears twice, the new one replaces the old one instead of piling up. Messages never cover the buttons of an open dialog.

A dialog with boxes to fill in or tick stays open if you click the dimmed area around it by mistake, so you do not lose your choices; its edge briefly lights up to show it is still waiting. To close it, click **Cancel** or press **Esc**. Dialogs that only show information close with a click outside.

The sample survey

The practice data is the "Urban Neighbourhoods and Social Trust Survey": 640 households in Kolkata, Delhi, Mumbai, Bengaluru and Chennai. It is invented for teaching, so no real person is described. You can load it again at any time with **File > Load sample survey**.

These are the variables this guide uses:

Variable	What it is
<code>gender</code>	1 = Man, 2 = Woman, 3 = Other / prefer to self-describe
<code>age</code>	Age in years (17 people did not give their age)
<code>educ</code>	Highest education, from 1 = No formal schooling to 6 = Postgraduate
<code>migrant</code>	0 = Born in this city, 1 = Moved here from another state or district
<code>yrs_nbhd</code>	Years lived in this neighbourhood
<code>trust1</code> to <code>trust5</code>	Five statements about trust in the neighbourhood, from 1 = Strongly disagree to 5 = Strongly agree
<code>life_sat</code>	Life satisfaction, from 0 to 10
<code>vote</code>	Voted in the last municipal election (0 = No, 1 = Yes)
<code>wt</code>	A survey weight
<code>q_challenge</code>	Open question: "What is the biggest challenge facing your neighbourhood today?"

CHAPTER 3

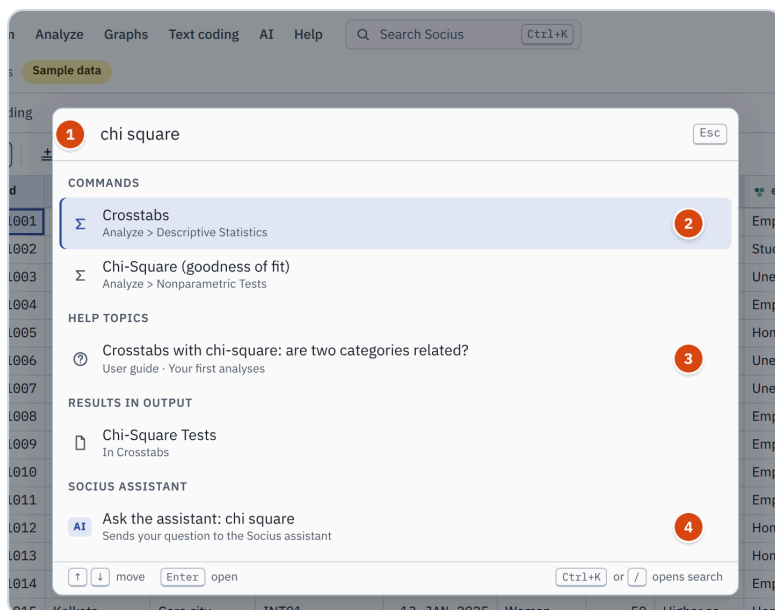
Finding anything with Search

You do not need to remember where things are in the menus. Search finds commands, variables, results and help pages for you.

Open Search

- Press `Ctrl+K` (`Cmd+K` on a Mac), from anywhere in Socius.
- Or press `/` when you are not typing in a box or in the Data View grid.
- Or click the **Search Socius** box in the top bar. On a narrow screen it becomes a magnifying-glass button, and on a phone the search opens full screen.

Start typing. Use the arrow keys to move through the results, `Enter` to open one and `Esc` to close Search. Small typing mistakes are fine: "crostabs" still finds Crosstabs.



Searching for "chi square". Type in the box (1). Commands (2) run the menu item, Help topics (3) open this guide at the right page, and the last line (4) asks the Socius assistant.

What you can search for

Type this	What you get
chi square	Crosstabs (under Analyze > Descriptive Statistics). Enter opens the dialog.
t test, anova, regression, alpha	The matching analysis. "alpha" or "reliability" finds Reliability Analysis.
recode, weight, filter	The matching Transform or Data command.
trust	Your variables whose name, label or value labels contain "trust". Enter selects its column in Data View; the Variable View and Frequencies buttons on each row open it there or in Frequencies.
the title of a result, such as Crosstabs	Results in Output: jumps to that result.
missing values	Help topics: opens this guide at that section, in a new tab.
a word from your interviews, such as water	Once you have texts in Text coding: Search in texts for "water" lists every passage that contains it (Keyword in context).
any question	Ask the assistant: ... (always the last line) sends your question to the Socius assistant.

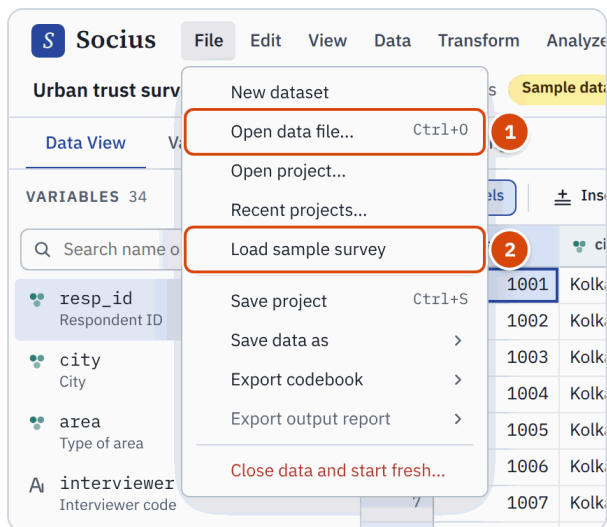
With an empty box, Search shows **Recent** (what you opened from Search lately) and **Suggestions**. Commands that cannot run yet are greyed out and say why, for example "Open or create a dataset first". If you choose one anyway, Search explains what is missing. When a command needs data, it offers two buttons: **Open data file...** and **Load sample survey**.

Greyed-out items in the menus work the same way: point at one to see why it is not available yet.

Opening your data

Open an SPSS file

1. Choose **File > Open data file** (or press `Ctrl+O`, `Cmd+O` on a Mac). You can also drag the file from your computer onto the Socius window.
2. Pick your `.sav` file (or a compressed `.zsav` file) and click **Open**.
3. A short message tells you how many cases and variables were read.



The File menu. Open data file (1) opens your own file; Load sample survey (2) brings back the practice data.

Everything that makes an SPSS file useful comes with it: variable labels (the full question text), value labels (for example 1 = Man, 2 = Woman), missing value codes, measurement levels and the weight variable.

Open a CSV or Excel file

Socius also opens CSV files (`.csv`, `.tsv`) and Excel workbooks (`.xlsx`). Use the same **File > Open data file**. Before the data opens, Socius shows a preview so you can check that the columns are split correctly.

Open my-survey.csv

Check how the text file is split into variables.

Separator

Detect automatically

Character encoding

Detect (UTF-8 if possible)

☒ First row has variable names

Start of the file

id,gender,age,city,satisfaction
1, Woman, 34, Kolkata, 7
2, Man, 51, Delhi, 6
3, Woman, 27, Mumbai, 8
4, Man, 45, Chennai, 5
5, Woman, 62, Kolkata, 9

Preview (5 variables, 5 cases)

id	gender	age	city	satisfaction
1	Woman	34	Kolkata	7
2	Man	51	Delhi	6
3	Woman	27	Mumbai	8
4	Man	45	Chennai	5
5	Woman	62	Kolkata	9

Cancel

Open

The preview for a CSV file. Check the separator, the character encoding and whether the first row holds variable names, then click Open.

- **Separator:** leave it on **Detect automatically** unless the preview looks wrong.
- **Character encoding:** if letters look strange (for example in Hindi, Bengali or accented names), try another encoding here.
- **First row has variable names:** keep this ticked if your first row holds column headings.
- For Excel files, you also choose which sheet to open.

CSV and Excel files do not store labels or missing value codes. After opening one, spend a few minutes in Variable View to add them (see the next chapter). If a column holds words, such as "Woman" and "Man", **Transform > Automatic recode** turns it into numbered categories with value labels, which is what most analyses need.

OPENING A FILE REPLACES THE OPEN DATA

Socius works on one dataset at a time. If you have unsaved changes, it asks before it replaces them.

CHAPTER 5

Understanding your variables

Before you analyse anything, spend five minutes in **Variable View**. It prevents many wrong results later. Click the **Variable View** tab.

	Name 1	Type	Width	Decimals	Label 2	Values 3	Missing 4	Columns	Align	Measure 5	Role
1	resp_id	Numeric	4	0	Respondent ID	None	None	6	Right	Nominal	Input
2	city	Numeric	1	0	City	1 = Kolkata +4	None	10	Right	Nominal	Input
3	area	Numeric	1	0	Type of area	1 = Core city +1	None	10	Right	Nominal	Input
4	interviewer	String	5		Interviewer code	None	None	8	Left	Nominal	Input
5	int_date	Date	11		Date of interview	None	None	11	Right	Scale	Input
6	gender	Numeric	1	0	Gender of respondent	1 = Man +2	None	8	Right	Nominal	Input
7	age	Numeric	3	0	Age in completed years	None	None	5	Right	Scale	Input
8	educ	Numeric	1	0	Highest level of education completed	1 = No formal schooling +5	None	8	Right	Ordinal	Input
9	employ	Numeric	1	0	Main employment status	1 = Employed full-time +6	None	8	Right	Nominal	Input
10	marital	Numeric	1	0	Marital status	1 = Never married +3	None	8	Right	Nominal	Input
11	hh_size	Numeric	2	0	Number of people usually living in the household	None	None	6	Right	Scale	Input
12	hh_income	Numeric	8	0	Total monthly household income from all source...	999999 = Refused	999999	9	Right	Scale	Input
13	migrant	Numeric	1	0	Were you born in this city, or did you move here f...	0 = Born in this city +1	None	8	Right	Nominal	Input
14	yrs_nbhd	Numeric	3	0	How many years have you lived in this neighbour...	None	None	6	Right	Scale	Input

Variable View. Each row describes one variable of your data.

The most important columns are:

- 1. Name.** A short name used in menus and output, such as `life_sat`.
- 2. Label.** The full description, often the question wording. Socius shows labels in dialogs and tables.
- 3. Values.** Value labels: what each code means, such as 1 = Man.
- 4. Missing.** Codes that mean "no real answer", such as 8 = Don't know or 9 = Refused.
- 5. Measure.** The measurement level: nominal, ordinal or scale.

Rename a variable

Short, clear names make your output easier to read. To rename a variable:

1. In Variable View, double-click its cell in the **Name** column (or click the cell and press `Enter` or `F2`, or just start typing).
2. Type the new name. You can click inside the box to fix a typo.
3. Press `Enter` or `Tab`, or click somewhere else.

In Data View, double-clicking a column heading takes you to Variable View with that variable's name ready to change.

The rules come from SPSS, so your file still opens there:

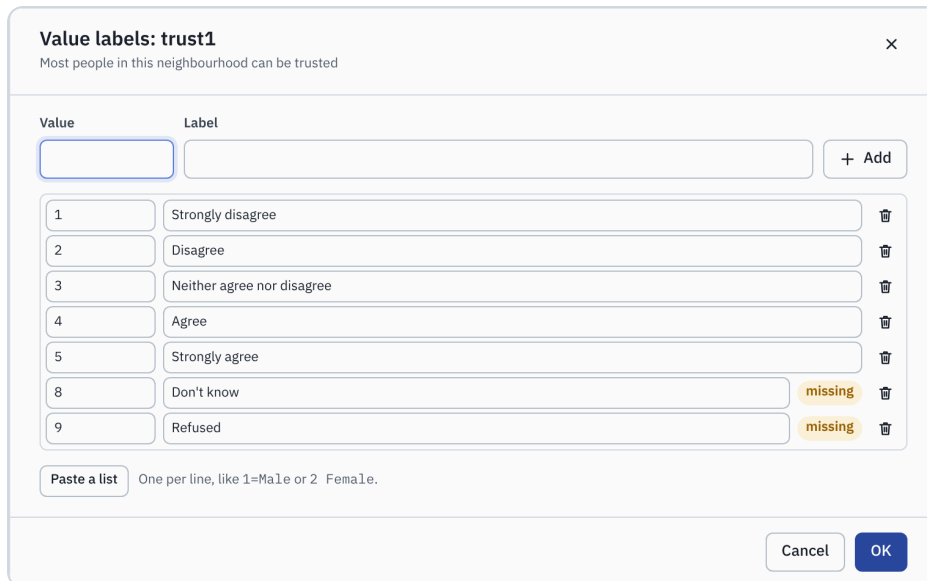
- Start with a letter. Letters from any alphabet work, so a Bengali name such as `বয়স` is fine.
- Use only letters, digits and `_ . @ # $`. No spaces: write `age_group`, not `age group`.
- Do not end with `.` or `_`.
- Every name must be different. Capitals do not count, so `age` and `AGE` are the same name. Changing only the capitals (`age` to `Age`) is fine.
- At most 64 bytes: 64 English letters, or about 21 Bengali letters.

If a name breaks a rule, a message under the cell says why and suggests a fix, for example "Names cannot contain spaces. Try `age_group`." Correct it and press `Enter` again, or press `Esc` to cancel. If you click elsewhere, the old name stays and a short message says why.

Renaming never breaks anything: weights, filters, exported codes and earlier results follow the variable, not its old name. **Edit > Undo** shows "Undo rename of" and the old name.

Value labels

Click a cell in the **Values** column and then the small ... button (or press Enter) to see or edit the value labels.



Value labels: trust1
Most people in this neighbourhood can be trusted

Value	Label	
		+ Add
1	Strongly disagree	🗑
2	Disagree	🗑
3	Neither agree nor disagree	🗑
4	Agree	🗑
5	Strongly agree	🗑
8	Don't know	missing 🗑
9	Refused	missing 🗑

Paste a list One per line, like 1=Male or 2 Female.

Cancel OK

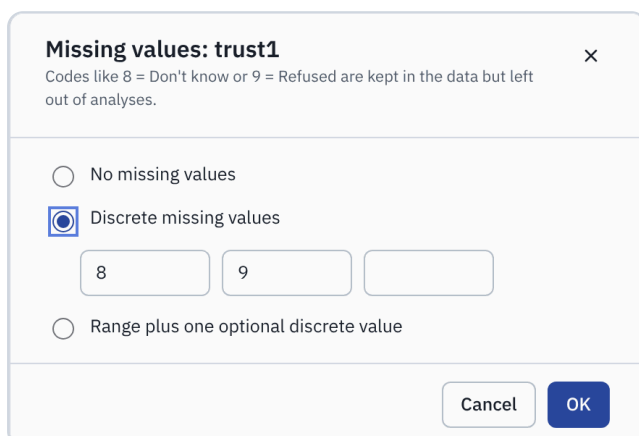
The value labels of trust1. Codes 8 and 9 are marked as missing.

To add a label, type the value and the label at the top and click **Add**. **Paste a list** lets you paste many labels at once, one per line, such as `1=Strongly disagree`.

Missing values

Surveys often use special codes for "Don't know" or "Refused". If you declare them as missing, Socius leaves them out of every percentage, mean and test. If you forget, a code like 9 would be treated as a real answer and pull your averages up.

Click a cell in the **Missing** column and then its ... button.



Missing values: trust1
Codes like 8 = Don't know or 9 = Refused are kept in the data but left out of analyses.

☐ No missing values

☒ Discrete missing values

☐ Range plus one optional discrete value

Cancel OK

The missing values of trust1: 8 and 9 are declared missing.

You can choose **No missing values**, **Discrete missing values** (up to three codes, such as 8 and 9), or **Range plus one optional discrete value** (for example 90 to 99).

DECLARE IT ONCE, USE IT EVERYWHERE

If many variables share the same coding, such as all items of a Likert scale, set the labels and missing values on one of them. Then use **Data > Copy variable properties** to copy them to the others. The next section shows a faster way to check many variables at once.

Check many variables at once: Define variable properties

Files from CSV or Excel often arrive without labels, and survey files often hide codes like 8 = Don't know or 99 = Refused among the real answers. **Data > Define variable properties** reads the values each variable really has and helps you label them, mark missing codes and set the measurement level, all in one place. It works like the SPSS command of the same name. You also find it on the Variable View toolbar and in the Variable View right-click menu.

Step 1: choose the variables. Tick the variables you want to check, for example `trust1` to `trust5` and `hh_income`, and click **Scan 6 variables** (the button counts them). Good first choices are questions from a CSV or Excel file and the items of a Likert scale. By default Socius scans all cases and shows at most 200 different values per variable; you can change both.

Step 2: check each variable. The list on the left (1) shows every scanned variable with its state, such as **Labels complete**, **3 values unlabelled** or **Suspected missing code**. Click a variable to see its values on the right.

Define variable properties

Step 2 of 2: check each variable. Nothing changes until you press Apply.

1

Scanned variables (6)

trust1

Labels complete

trust2

trust3

trust4

trust5

hh_income

trust1

Numeric (F1.0)

Variable label (the question wording)

Most people in this neighbourhood can be trusted

Measurement level

Ordinal

Ordinal suggested: 5 ordered codes with labels like Strongly disagree ... Strongly agree. (the current setting)

Width

1

Decimals

0

Valid 626

Missing 14 (14 marked missing, 0 empty)

Unique values 7

Cases scanned 640

Suggest labels

Copy properties from another variable...

Apply these properties to other variables...

Missing values: 8, 9

Tick Missing for codes like Don't know or Refused: they stay in the data but analyses leave them out. SPSS allows 3 single missing values, or a range plus 1.

Value	Label	Missing	Count	Notes
1	Strongly disagree	☐	49	
2	Disagree	☐	132	
3	Neither agree nor disagree	☐	182	
4	Agree	☐	184	
5	Strongly agree	☐	79	
7	Not applicable	☐	0	looks like a missing code
8	Don't know	☒	13	missing code
9	Refused	☒	1	missing code
Value	Label for a value not in the data	+ Add		

Choose variables

Cancel

Apply (1 variable)

Checking `trust1`. The scanned variables (1), Socius's suggestion for the measurement level (2), codes already marked missing (3), a code that looks like a missing code (4) and Apply (5).

- **Measurement level.** Socius suggests a level and says why (2), for example "Ordinal suggested: 5 ordered codes with labels like Strongly disagree ... Strongly agree." If the suggestion differs from the current level, a button such as **Use Ordinal** switches it.
- **The value grid.** Every value in the data, with its **Label**, a **Missing** box and a **Count**. Type a label straight into the grid. To label a code that does not occur in the data (yet), type it in the last row, **Label for a value not in the data**, and click **Add**.
- **Notes.** **missing code** (green, 3) means the code is already marked missing. **looks like a missing code** (amber, 4) flags a code such as 7, 8 or 9 on a 1 to 5 scale, 98, 99, 999, negative codes like -1, or a label such as "Don't know" or "Refused". Tick **Missing** if it is not a real answer. **unlabelled** marks a value with no label.
- **Suggest labels** offers ready-made labels that fit the values, such as an agreement scale from 1 to 5, yes/no, or labels for the missing codes 8 and 9. You see a preview before anything changes. Always check them against your questionnaire.
- **Copy properties from another variable...** and **Apply these properties to other variables...** copy the labels, missing values and level between variables. For a battery such as `trust1` to `trust5`, one click selects the other items.

Nothing changes until you click **Apply** (5). All your edits then become one change: **Edit > Undo** in Data View or Variable View takes all of them back at once. The SPSS syntax is written to Output. If you click **Cancel** after editing, Socius asks whether to discard your changes.

SPSS LIMITS, EXPLAINED ON THE SPOT

SPSS allows at most three single missing values per variable (or a range plus one value), and limits labels to 120 bytes. If you go over a limit, Socius says so next to the value, and **Apply** waits until you fix it.

Measurement level: nominal, ordinal or scale

The measurement level tells Socius what kind of information a variable holds. Socius uses it to suggest suitable analyses and to warn you when a variable does not fit a box in a dialog.

- **Nominal:** categories with no order. Examples: gender, city, religion. You can count them, but "more" or "less" makes no sense.
- **Ordinal:** categories with a clear order, but the steps are not necessarily equal. Examples: education level, "Strongly disagree" to "Strongly agree".
- **Scale:** real numbers where differences are meaningful. Examples: age in years, income, a 0 to 10 satisfaction score.

Small icons next to each variable show its level: a cluster of circles for nominal, rising bars for ordinal and a ruler for scale. To change a level, click the **Measure** cell and pick another.

Preparing your data

Real data rarely arrives ready to analyse. The **Transform** and **Data** menus help you reshape it. Each example here uses the sample survey. Every change is written to the Output tab together with the SPSS command that does the same thing, and **Edit > Undo** (**Ctrl+Z**) in Data View or Variable View reverses it. When a command creates a new variable, the message that appears offers **Show**, which selects the new column in Data View.

Recode age into groups

Goal: a new variable `agegrp` with four age groups, keeping the original `age`.

1. Choose **Transform > Recode into different variables**.
2. Tick `age` in the list on the left.
3. Under **New variables**, type the name `agegrp` and the label `Age group`.
4. Add the rules on the right, one at a time. For each rule, choose the kind of old value, type the numbers, type the new value and click **Add**:
 - **System- or user-missing** becomes **System-missing** (so missing ages stay missing)
 - **Lowest through 29** becomes 1
 - **Range 30 to 44** becomes 2
 - **Range 45 to 64** becomes 3
 - **through Highest 65** becomes 4
5. Under **Labels for the new codes**, type 18-29, 30-44, 45-64 and 65 and over.
6. Click **OK**.

Recode into Different Variables
Make new variables from old codes, for example age groups from age. The original variables stay as they are.

Variables to recode

Search variables: All None

- ☐ city City
- ☐ area Type of area
- ☐ interviewer Interviewer code
- ☐ int_date Date of interview
- ☐ gender Gender of respondent
- ☒ age Age in completed years 1

1 selected

New variables

age > agegrp Age group

☒ Numeric result ☐ Text result, width 8

Old and new values

Old value: through Highest value

New value: Value value + Add

System- or user-missing	>	System-missing	↑	↓	⌵
Lowest thru 29	>	1	↑	↓	⌵
30 thru 44	>	2	↑	↓	⌵
45 thru 64	>	3	↑	↓	⌵
65 thru Highest	>	4	↑	↓	⌵

Values without a rule become missing in the new variable.
Keep them: add "All other values = Copy old value"

Labels for the new codes (optional)

1 18-29

2 30-44

3 45-64

4 65 and over

☐ Only recode cases that meet a condition (if)

Cancel OK

Recoding age into four groups. The rules are checked from top to bottom, and the first rule that fits is used.

The new variable appears at the end of your data. Values without a rule become missing in the new variable, so check that your rules cover every age. If no rule says what to do with **All other values**, the dialog warns you in an orange box. When you want to keep every other value as it is, click **Keep them: add "All other values → Copy old value"** in that box, and Socius adds the rule for you. In this example the rules already cover every age (and missing ages stay missing), so you can leave the warning as it is.

Reverse-code a question

Four of the trust statements are positive ("Most people in this neighbourhood can be trusted"). One is negative: `trust3` says "You have to be very careful with people in this neighbourhood". Agreeing with it means *less* trust. Before you combine the five items, flip `trust3` so that a high score always means more trust.

1. Choose **Transform > Reverse-code items**.
2. Tick `trust3`. Socius reads the answer scale (1 to 5) from the value labels.
3. Keep **Create new variables, named with the suffix `_r`** and click **OK**.

You now have `trust3_r`. Its value labels are flipped too, and the missing codes 8 and 9 stay as they are.

Build a scale and check Cronbach's alpha

A scale (or index) combines several questions about the same idea into one score. It is usually more reliable than any single question.

1. Choose **Transform > Create scale / index**.
2. Tick `trust1`, `trust2`, `trust3_r`, `trust4` and `trust5`. Use `trust3_r`, not `trust3`.
3. Type the name `trust` and the label `Neighbourhood trust (mean of 5 items)`.
4. Keep **Mean of the answered items**. The score then stays on the familiar 1 to 5 scale.
5. **Minimum answered items** is 4: a person who answered fewer than four of the five gets no score.
6. Click **OK**.

Creating a trust scale. The box (1) shows Cronbach's alpha for the items you picked.

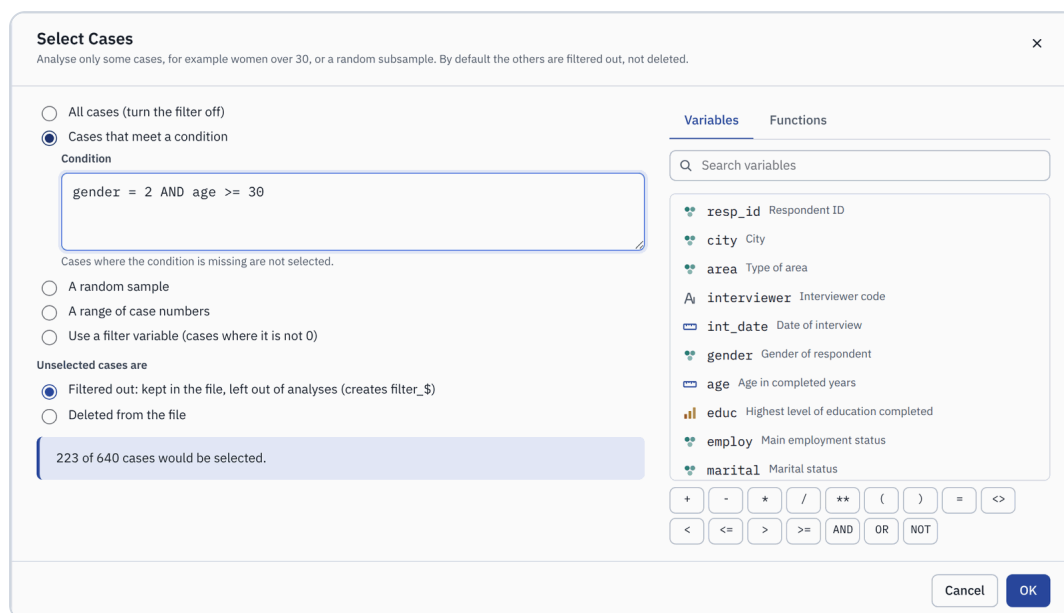
What Cronbach's alpha means. Alpha tells you how well the items hang together, that is, whether people who agree with one tend to agree with the others. It runs from 0 to 1. As a rough guide, .70 or higher is acceptable for a research scale and .80 or higher is good. Here alpha is .78, so the five items form an acceptable scale. If you had used the original `trust3` by mistake, alpha would drop and Socius would warn you that the item "goes against the other items".

For the full picture, run **Analyze > Scale > Reliability Analysis** with the same five items.

Select cases

Sometimes you want to analyse only part of your sample, for example only women aged 30 and over.

1. Choose **Data > Select cases**.
2. Choose **Cases that meet a condition** and type the condition: `gender = 2 AND age >= 30`.
3. Keep **Filtered out** for unselected cases. They stay in the file but are left out of analyses.
4. Click **OK**.



Selecting women aged 30 and over. The box below the condition says how many cases would be selected.

While a filter is on, the dataset bar shows **Filter on** with the number of cases in use. To use everyone again, choose **Data > Turn filter off (use all cases)**, or click the small cross on that label.

Weight cases

Many surveys come with a weight that corrects for how the sample was drawn. The sample survey has one called `wt`.

1. Choose **Data > Weight cases**.
2. Choose **Weight cases by** and pick `wt`.
3. Click **OK**.

The dataset bar now says **Weighted by wt**, and every analysis uses the weight until you switch it off with **Data > Turn weighting off** (or the cross on the label).

The label in the dataset bar shows that a weight is on (1).

NOTE

The rest of this guide uses the unweighted data, so switch weighting off again if you tried it. Each result in Output says whether cases were weighted.

CHAPTER 7

Your first analyses

All analyses live in the **Analyze** menu, and they all work the same way:

1. Pick an analysis from the menu. A dialog opens.
2. Move variables from the list on the left into the boxes on the right. Double-click a variable, drag it, or select it and click the arrow.
3. Check the options below the boxes, if you need to.
4. Click **Run**. The result appears in the **Output** tab.

Every analysis dialog has a **When to use this** link at the top that explains in plain words what the analysis is for.

LARGE DATASETS

Most analyses finish in a moment. A large one (roughly 50,000 cells or more, for example 5,000 cases and 10 variables) runs in the background, so Socius stays usable. The dialog then shows a moving bar with "Running in the background" and the seconds so far, and **Cancel** becomes **Stop**. Click **Stop** if you want to change something first.

Every result comes with two extra boxes. **What this means** explains the result in plain language, to help you read it. **APA-style report** gives you a draft sentence for your results section, written in the style of the American Psychological Association (APA 7), which most sociology journals and departments accept.

Frequencies: how often does each answer occur?

When to use it. To describe one variable at a time: how many people gave each answer, and what percentage that is. It is usually the first thing you run on any survey.

Try it. How educated is the sample?

1. Choose **Analyze > Descriptive Statistics > Frequencies**.
2. Find **educ** in the list (1) and move it into **Variable(s)** (3) with the arrow (2).
3. Click **Run** (4).

Frequencies

Count how often each answer occurs, with percentages, summary statistics and charts.

When to use this

- Respondent ID [resp_id]
- City [city]
- Type of area [area]
- Interviewer code [interviewer]
- Date of interview [int_date]
- Gender of respondent [gender]
- Age in completed years [age]
- Highest level of education completed [educ] ✓
- Main employment status [employ]
- Marital status [marital]
- Number of people usually living in the household [hh_size]
- Total monthly household income from all sources (INR) [hh_i...
- Were you born in this city, or did you move here from another ...
- How many years have you lived in this neighbourhood? (years)

→

Variable(s)

1 or more

Highest level of education completed [educ] X

Tables

Statistics

Charts

☒ Display frequency tables

Order categories by

Ascending values

Show syntax

Reset

Cancel

Run

The Frequencies dialog with educ in the Variable(s) box.

Frequencies

Urban trust survey (sample) · 24 Sep 2026, 18:41 · N = 640.

Copy

AI Explain with AI

↑

↓

🗑️

Table 1

Statistics

		educ
N	Valid	640
	Missing	0

Copy table

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Table 2

Highest level of education completed

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No formal schooling	40	6.3	6.3	6.3
	Primary	64	10.0	10.0	16.3
	Secondary	158	24.7	24.7	40.9
	Higher secondary	148	23.1	23.1	64.1
	Graduate	152	23.8	23.8	87.8
	Postgraduate	78	12.2	12.2	100.0
Total		640	100.0	100.0	

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WHAT THIS MEANS

For Highest level of education completed, the most common answer was "Secondary" (24.7% of 640 valid responses), followed by "Graduate" (23.8%). The least common was "No formal schooling" (6.3%).

APA-STYLE REPORT

Copy

Responses to Highest level of education completed (N = 640) were distributed as follows: No formal schooling: 6.3% (n = 40); Primary: 10.0% (n = 64); Secondary: 24.7% (n = 158); Higher secondary: 23.1% (n = 148); Graduate: 23.8% (n = 152); Postgraduate: 12.2% (n = 78).

The result: a frequency table for education, a plain-language summary and a draft APA sentence.

How to read it.

- **Frequency** is the number of people who gave each answer.
- **Percent** is the share of all cases, including missing ones.
- **Valid Percent** leaves out missing answers. This is usually the number to report.

- **Cumulative Percent** adds the valid percentages up as you go down the table. Here 40.9% have secondary education or less.

The most common answer is "Secondary" (24.7%), and the least common is "No formal schooling" (6.3%).

HOW TO REPORT IT

Of the 640 respondents, 24.7% had completed secondary school, 23.8% held a graduate degree and 12.2% a postgraduate degree; 6.3% had no formal schooling.

Crosstabs with chi-square: are two categories related?

When to use it. When both variables are categories (nominal or ordinal), and you want to know whether they are related. For example: do women and men feel equally safe in their neighbourhood?

Try it. Does the feeling of safety after dark differ by gender?

1. Choose **Analyze > Descriptive Statistics > Crosstabs**.
2. Put **gender** in **Row(s)** and **trust5** ("I would feel safe walking alone here after dark") in **Column(s)**.
3. On the **Cells** tab, keep **Row percentages** ticked. Put the variable that defines your groups in the rows; then each row adds up to 100% and you compare rows.
4. On the **Statistics** tab, **Chi-square** and **Phi and Cramer's V** are already ticked.
5. Click **Run**.

Crosstabs
Cross-tabulate two categorical variables, test whether they are associated, and add a control variable to elaborate the relationship.

When to use this

Search 37 variables

- Respondent ID [resp_id]
- City [city]
- Type of area [area]
- Interviewer code [interviewer]
- Date of interview [int_date]
- Gender of respondent [gender]
- Age in completed years [age]
- Highest level of education completed [educ]
- Main employment status [employ]
- Marital status [marital]
- Number of people usually living in the household [hh_size]
- Total monthly household income from all sources (INR) [hh_1...
- Were you born in this city, or did you move here from another ...

Row(s) 1 or more - nominal or ordinal
Gender of respondent [gender]

Column(s) 1 or more - nominal or ordinal
I would feel safe walking alone here after dark [trust5]

Layer (control variable) Optional, 1 variable - nominal or ordinal
Optional: produces a table per category of the control variable plus the total.

Cells Statistics Exact Charts

☒ Chi-square ☒ Phi and Cramer's V ☐ Contingency coefficient ☐ Lambda and Goodman-Kruskal tau

☐ Gamma ☐ Somers' d ☐ Kendall's tau-b ☐ Kendall's tau-c

☐ Correlations (Spearman, Pearson) ☐ Kappa ☐ Risk (odds ratio, 2x2) ☐ McNemar

☐ Cochran-Mantel-Haenszel (layered 2x2)

Show syntax Reset Cancel Run

The Crosstabs dialog, Statistics tab. Chi-square and Cramer's V are ticked by default.

Table 4*gender * trust5 Crosstabulation*

Gender of respondent by I would feel safe walking alone here after dark

			trust5					
			Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree	Total
gender	Man	Count	17	72	88	105	39	321
		% within gender	5.3%	22.4%	27.4%	32.7%	12.1%	100.0%
	Woman	Count	42	86	77	61	24	290
		% within gender	14.5%	29.7%	26.6%	21.0%	8.3%	100.0%
	Other / prefer to self-describe	Count	5	4	6	1	3	19
		% within gender	26.3%	21.1%	31.6%	5.3%	15.8%	100.0%
Total		Count	64	162	171	167	66	630
		% within gender	10.2%	25.7%	27.1%	26.5%	10.5%	100.0%

The crosstabulation: counts and row percentages for each gender.

How to read the table. Compare the percentages across each row. Among men, 32.7% agree and 12.1% strongly agree that they feel safe after dark: 44.8% in total. Among women the total is 29.3% (21.0% plus 8.3%). Women are also more likely to strongly disagree (14.5% against 5.3% of men).

Table 5*Chi-Square Tests*

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	35.016 ^a	8	< .001
Likelihood Ratio	35.989	8	< .001
Linear-by-Linear Association	24.100	1	< .001
N of Valid Cases	630		

Note. a. 3 cells (20.0%) have expected count less than 5. The minimum expected count is 1.93.

 Copy table  Excel

Table 6*Symmetric Measures*

		Value	Asymptotic Standard Error	Approximate T	Approximate Significance
Nominal by Nominal	Phi	.236			< .001
	Cramer's V	.167			< .001
N of Valid Cases		630			

The chi-square test and the strength of the association (Cramer's V).

How to read the test.

- **Pearson Chi-Square** asks: could a difference this large appear by chance if gender and feeling safe were unrelated? The **p-value** (in the "Significance" column) answers it. Here $p < .001$, far below the usual .05 limit, so the difference is very unlikely to be chance. We call it *statistically significant*.
- **Cramer's V** tells you how *strong* the relationship is, from 0 (none) to 1 (perfect). As a rough guide, .10 is weak, .30 moderate and .50 strong. Here $V = .17$: a real but weak relationship.
- The note under the table warns you if many cells have very few people ("expected count less than 5"). If more than 20% of cells are affected, the test becomes unreliable; merge small categories or use the **Exact** tab.

WHAT THIS MEANS

Respondents in the "Man" group of Gender of respondent were more likely to answer "Agree" on I would feel safe walking alone here after dark (32.7%) than those in the "Other / prefer to self-describe" group (5.3%). The association is statistically significant and weak (Cramer's $V = .17$; $p < .001$). The cells that differ most from independence are: fewer "Man" respondents answering "Strongly disagree" than expected (adjusted residual -4.1); more "Man" respondents answering "Agree" than expected (adjusted residual +3.6); more "Woman" respondents answering "Strongly disagree" than expected (adjusted residual +3.3). Effect-size labels follow Cohen (1988) conventions; judge practical importance in the context of your research question.

APA-STYLE REPORT

 Copy

A chi-square test of independence showed a significant association between Gender of respondent and I would feel safe walking alone here after dark, $\chi^2(8, N = 630) = 35.02, p < .001$, Cramér's $V = .17$.

Socius explains the result in plain words and drafts the APA sentence.

HOW TO REPORT IT

A chi-square test of independence showed a significant association between gender and feeling safe walking alone after dark, $\chi^2(8, N = 630) = 35.02, p < .001$, Cramér's $V = .17$. Men were more likely than women to agree that they felt safe (44.8% vs. 29.3%).

Click **Copy** in the APA box to copy the sentence. Before you paste it into your paper, replace the long question wording with a short description, and put the statistical symbols (χ^2 , N , p , V) in italics.

Comparing two groups: the independent-samples t-test

When to use it. When you compare the *average* of a scale variable between exactly two groups. For example: are people who were born in the city more satisfied with life than people who moved there?

Try it.

1. Choose **Analyze > Compare Means > Independent-Samples T Test**.
2. Put `life_sat` in **Test Variable(s)** and `migrant` in **Grouping Variable**.
3. Under **Groups**, Socius fills in the two values of `migrant`: "Born in this city (0)" and "Migrated from another state or district (1)".
4. Click **Run**.

Table 7

Group Statistics

	migrant	N	Mean	Std. Deviation	Std. Error Mean
All things considered, how satisfied are you with your life as a whole these days? (0-10)	Born in this city	366	6.37	1.80	0.094
	Migrated from another state or district	264	5.70	1.85	0.114

[Copy table](#)
[Excel](#)
Table 8

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
										95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
All things considered, how satisfied are you with your life as a whole these days? (0-10)	Equal variances assumed	0.282	.595	4.506	628	< .001	< .001	0.66	0.147	0.37	0.95
	Equal variances not assumed			4.485	556.898	< .001	< .001	0.66	0.148	0.37	0.95

Note. Levene's test is based on the mean (absolute deviations from each group mean), as in SPSS.

[Copy table](#)
[Excel](#)

Group means and the t-test.

How to read it.

- **Group Statistics** shows the mean (average) for each group. People born in the city score 6.37 on average; migrants score 5.70.
- **Levene's Test** checks whether the two groups are equally spread out. If its "Sig." is above .05 (here .595), read the row **Equal variances assumed**. If it is below .05, read **Equal variances not assumed**. The plain-language box tells you which row to read.
- **t**, **df** and **Two-Sided p** give the test. Here $t(628) = 4.51$, $p < .001$: the difference is statistically significant.
- **Cohen's d** (in the effect sizes table) tells you how big the difference is: about 0.2 is small, 0.5 medium and 0.8 large. Here $d = 0.36$, a small to medium difference.

HOW TO REPORT IT

Respondents born in the city reported higher life satisfaction ($M = 6.37$, $SD = 1.80$) than migrants ($M = 5.70$, $SD = 1.85$), $t(628) = 4.51$, $p < .001$, $d = 0.36$.

Comparing several groups: one-way ANOVA

When to use it. Like the t-test, but for three or more groups. For example: does life satisfaction differ by level of education?

Try it.

1. Choose **Analyze > Compare Means > One-Way ANOVA**.

- Put **life_sat** in **Dependent List** and **educ** in **Factor**.
- On the **Post hoc** tab, tick **Tukey**. This compares every pair of groups.
- Click **Run**.

Table 10

Descriptives

All things considered, how satisfied are you with your life as a whole these days? (0-10)

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
No formal schooling	39	4.95	1.90	0.305	4.33	5.57	1	9
Primary	63	5.52	1.79	0.226	5.07	5.98	0	9
Secondary	157	5.97	1.82	0.146	5.68	6.26	2	10
Higher secondary	145	6.14	1.83	0.152	5.85	6.44	1	10
Graduate	151	6.24	1.73	0.141	5.96	6.52	3	10
Postgraduate	75	7.00	1.71	0.197	6.61	7.39	1	10
Total	630	6.09	1.85	0.074	5.94	6.23	0	10

Copy table Excel

Table 11

Tests of Homogeneity of Variances

All things considered, how satisfied are you with your life as a whole these days? (0-10)

	Levene Statistic	df1	df2	Sig.
Based on Mean	0.503	5	624	.774
Based on Median	0.443	5	624	.819

Copy table Excel

Table 12

ANOVA

All things considered, how satisfied are you with your life as a whole these days? (0-10)

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	139.194	5	27.839	8.669	< .001
Within Groups	2,003.828	624	3.211		
Total	2,143.022	629			

Copy table Excel

Descriptives, Levene's test and the ANOVA table.

How to read it.

- Descriptives** lists the mean for each group. Life satisfaction rises with education, from 4.95 (no formal schooling) to 7.00 (postgraduate).
- ANOVA** tests whether *any* of the group means differ. Here $F(5, 624) = 8.67$, $p < .001$, so they do. It does not say which groups differ.
- ANOVA Effect Sizes** gives eta squared (η^2): the share of the differences in life satisfaction that education accounts for. Here $\eta^2 = .065$, about 6.5%. As a rough guide, .01 is small, .06 medium and .14 large.
- Multiple Comparisons** (the Tukey test) shows every pair of groups. A star (*) marks pairs that differ significantly, for example postgraduates and graduates.
- If Levene's test is significant (below .05), the groups are unequally spread. Then rely on the **Welch** test and choose **Games-Howell** instead of Tukey.

HOW TO REPORT IT

A one-way ANOVA showed that life satisfaction differed significantly by education, $F(5, 624) = 8.67$, $p < .001$, $\eta^2 = .06$. Tukey post hoc tests showed that postgraduates ($M = 7.00$) were more satisfied than every other group.

Correlation: do two numbers move together?

When to use it. When both variables are scale (or ordinal) and you want to know whether higher values of one go with higher (or lower) values of the other.

Try it. Choose **Analyze > Correlate > Bivariate Correlations**, move `age`, `yrs_nbhd`, `life_sat` and `trust` (the scale you built) into **Variables**, and click **Run**.

Table 16
Correlations

		Age in completed years	How many years have you lived in this neighbourhood?	All things considered, how satisfied are you with your life as a whole these days? (0-10)	Neighbourhood trust (mean of 5 items)
Age in completed years	Pearson Correlation	1.000	.550**	-.126**	.188**
	Sig. (2-tailed)		< .001	.002	< .001
	N	623	623	613	622
How many years have you lived in this neighbourhood?	Pearson Correlation	.550**	1.000	.083*	.369**
	Sig. (2-tailed)	< .001		.038	< .001
	N	623	640	630	639
All things considered, how satisfied are you with your life as a whole these days? (0-10)	Pearson Correlation	-.126**	.083*	1.000	.289**
	Sig. (2-tailed)	.002	.038		< .001
	N	613	630	630	629
Neighbourhood trust (mean of 5 items)	Pearson Correlation	.188**	.369**	.289**	1.000
	Sig. (2-tailed)	< .001	< .001	< .001	
	N	622	639	629	639

Note. **. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

A correlation matrix. Each cell shows Pearson's r , its significance and the number of cases.

How to read it. Each cell pairs two variables. The **Pearson Correlation** (r) runs from -1 to +1:

- A positive r means both rise together; a negative r means one rises as the other falls.
- As a rough guide, .10 is weak, .30 moderate and .50 strong.
- Stars mark significant correlations.

Here, people who have lived longer in the neighbourhood trust their neighbours more ($r = .37$, moderate), and people with more trust are more satisfied with life ($r = .29$, weak). Remember that a correlation does not show which causes which.

HOW TO REPORT IT

Neighbourhood trust was positively correlated with years lived in the neighbourhood, $r(637) = .37, p < .001$, and with life satisfaction, $r(627) = .29, p < .001$.

ORDINAL VARIABLES

For a single Likert item or other ordered categories, also tick **Spearman** in the dialog. It uses ranks and does not assume equal steps between answers.

A simple linear regression

When to use it. To predict a scale outcome from one or more other variables, and to see how much the outcome changes for each one-unit change in a predictor.

Try it. How much does life satisfaction rise with neighbourhood trust?

1. Choose **Analyze > Regression > Linear Regression**.
2. Put `life_sat` in **Dependent** and `trust` in **Independent(s): Block 1**.
3. Click **Run**.

Table 18

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.289 ^a	.083	.082	1.770

Note. a. Predictors: (Constant), trust

Dependent Variable: All things considered, how satisfied are you with your life as a whole these days? (0-10) (life_sat)

 Copy table  Excel

Table 19

ANOVA

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	178.884	1	178.884	57.104	< .001
	Residual	1,964.130	627	3.133		
	Total	2,143.014	628			

Note. Dependent Variable: All things considered, how satisfied are you with your life as a whole these days? (0-10) (life_sat)

Model 1 predictors: (Constant), trust

 Copy table  Excel

Table 20

Coefficients

Model		Unstandardized Coefficients		Standardized Coefficients		Sig.	95% Confidence Interval for B		Collinearity Statistics	
		B	Std. Error	Beta	t		Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	4.217	0.258		16.376	< .001	3.712	4.723		
	trust	0.631	0.083	0.289	7.557	< .001	0.467	0.794	1.000	1.000

Note. Dependent Variable: All things considered, how satisfied are you with your life as a whole these days? (0-10) (life_sat)

 Copy table  Excel

Model Summary, ANOVA and Coefficients tables.

How to read it.

- **Model Summary: R Square** is the share of the differences in the outcome that the model explains. Here $R^2 = .083$: trust explains about 8% of the differences in life satisfaction.
- **ANOVA** tests the model as a whole. Here it is significant ($p < .001$).
- **Coefficients: B** is the key number. For each one-point rise in trust (on its 1 to 5 scale), life satisfaction rises by 0.63 points (on its 0 to 10 scale). The **(Constant)** is the predicted value when the predictor is zero, and is rarely interpreted.
- **Beta** is the same effect in standard units, useful for comparing predictors measured in different units.

HOW TO REPORT IT

A simple linear regression showed that neighbourhood trust predicted life satisfaction, $B = 0.63$, 95% CI [0.47, 0.79], $\beta = .29$, $t(627) = 7.56$, $p < .001$. The model explained 8.3% of the variance, $F(1, 627) = 57.10$, $p < .001$.

With one predictor Socius calls it a "simple" linear regression; with two or more, a "multiple" linear regression.

You can add more predictors to Block 1. Categorical predictors such as [gender](#) or [educ](#) are turned into 0/1 "dummy" variables for you, each compared with the first category.

Where to go next

Once you are comfortable with the analyses above, these are the natural next steps. They work the same way: pick variables, click **Run**, and read the plain-language box and the APA sentence.

- **Binary logistic regression (Analyze > Regression > Binary Logistic Regression)** predicts a yes/no outcome, such as whether someone voted. It reports odds ratios.
- **Reliability analysis (Analyze > Scale > Reliability Analysis)** gives Cronbach's alpha with item statistics, so you can see which question fits a scale worst.
- **Factor analysis (Analyze > Dimension Reduction > Factor Analysis)** shows which questions group together, for example whether trust items and civic activity items measure two different things.
- **Nonparametric tests (Analyze > Nonparametric Tests)** such as Mann-Whitney U and Kruskal-Wallis are alternatives to the t-test and ANOVA for ordinal or very skewed data.

CHAPTER 8

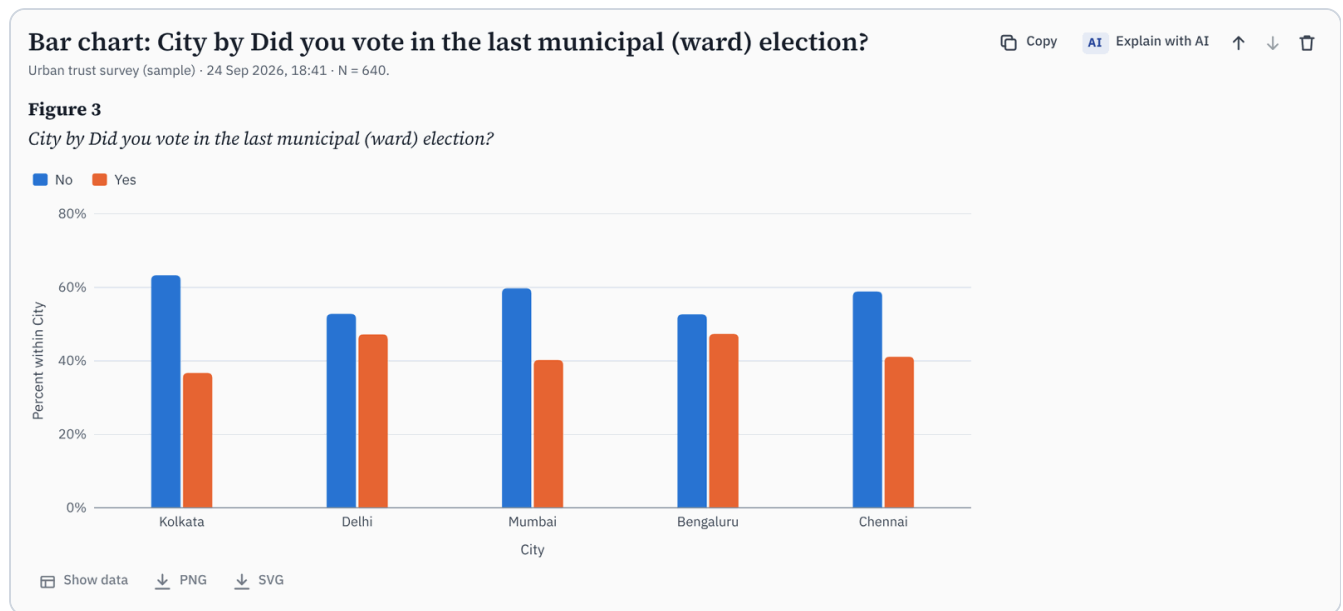
Charts

All charts are in the **Graphs** menu. They appear in Output with a title and the numbers behind them. In APA style (the standard setting), each chart gets a number in bold, such as **Figure 3**, with its title in italics above it, just as tables are numbered Table 1, Table 2 and so on. Numbers count up through the whole Output tab, so charts that other analyses draw (such as the residual plots of a regression) count too.

A bar chart

Try it. Did the share of voters differ between cities?

1. Choose **Graphs > Bar Chart**.
2. Put **city** in **Category axis** and **vote** in **Cluster by (optional)**.
3. Set **Bars show** to **Percent of cases** and **Percentages within** to **Each category**. Each city's bars then add up to 100%.
4. Click **Run**.

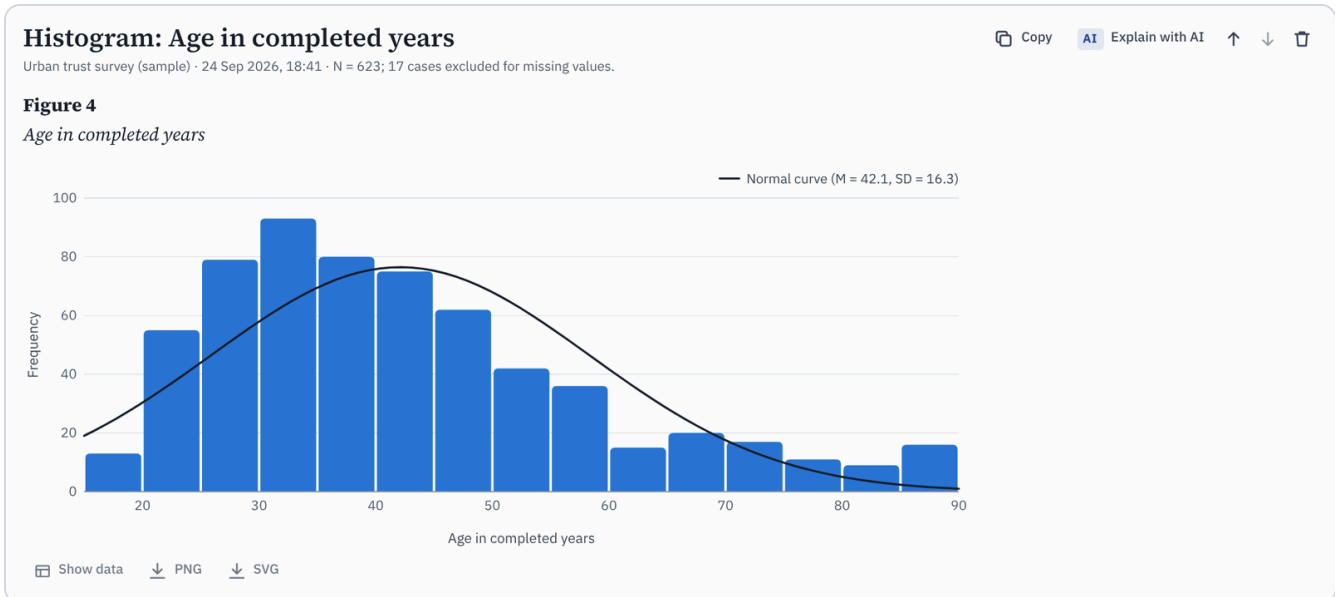


Voting by city, numbered Figure 3 in APA style. In every city, fewer than half of respondents voted.

For a single variable, leave **Cluster by** empty. To compare an average, such as mean trust by city, set **Bars show** to **Mean of a variable** and put the scale variable in **Variable for means**.

A histogram

A histogram shows the shape of a scale variable. Choose **Graphs > Histogram**, put **age** in **Variable** and click **Run**.



The age distribution with a normal curve. The buttons below the chart (1) show the data or save the chart as a picture.

Here the middle half of respondents are between 30 and 51 years old, with a long tail of older people. Socius calls this "right-skewed".

Saving a chart

Under every chart you find three buttons:

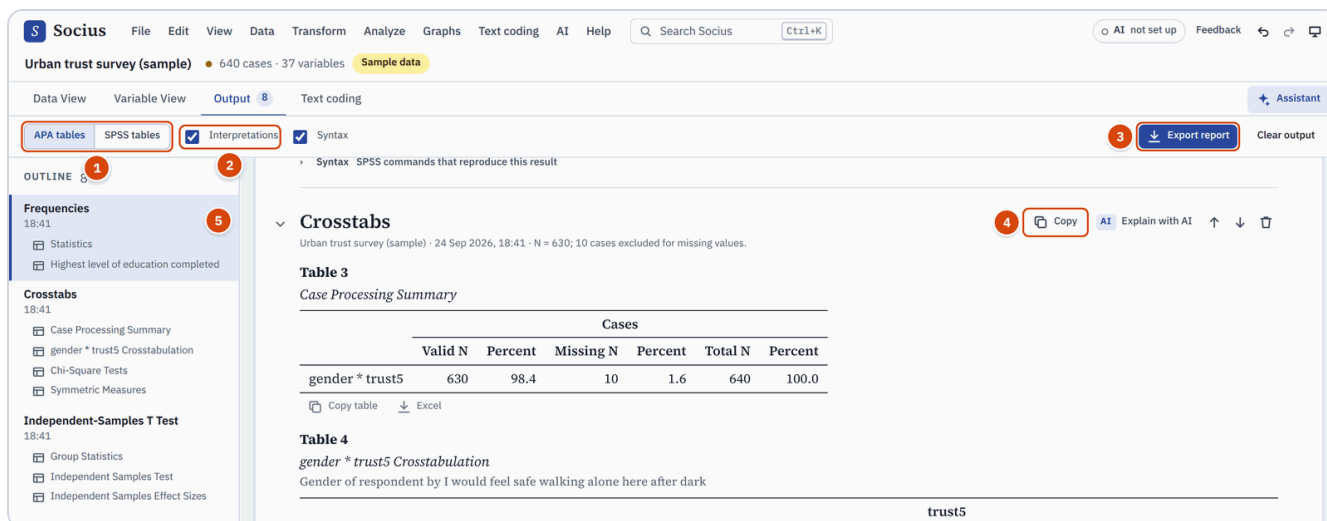
- **Show data** lists the numbers behind the chart.
- **PNG** saves a picture you can insert in Word or PowerPoint.
- **SVG** saves a sharp vector version that stays crisp at any size, useful for publications.

A PNG or SVG file keeps the chart's title inside the picture, so it still makes sense on its own. Charts are also included when you export the whole report to Word or as a web page (see the next chapter); in APA style they keep their "Figure" number and italic title there, ready for your thesis.

CHAPTER 9

Getting results into your report

Every analysis you run is added to the **Output** tab, newest at the bottom. The bar at the top of Output controls how results look and how you take them away.



The Output tab. The numbers are explained below.

1. **APA tables / SPSS tables** switches the table style.
2. **Interpretations** and **Syntax** show or hide the plain-language boxes and the SPSS commands.
3. **Export report** saves all the output in one file.
4. **Copy** at the top right of each result copies that result. Next to it, **Explain with AI** asks the optional AI help to explain the result (see Explain a result (page 49)).
5. The **Outline** lists every result, so you can jump between them.

APA style or SPSS style

APA tables have horizontal lines only, a numbered title and an italic heading. Charts get a matching "Figure" number. This is what most journals and theses expect. **SPSS tables** look like the SPSS viewer, which helps when you compare your results with SPSS or with a colleague who uses it; in SPSS style a chart keeps its title inside the chart instead of a Figure number.

Table 2

Highest level of education completed

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No formal schooling	40	6.3	6.3	6.3
	Primary	64	10.0	10.0	16.3
	Secondary	158	24.7	24.7	40.9
	Higher secondary	148	23.1	23.1	64.1
	Graduate	152	23.8	23.8	87.8
	Postgraduate	78	12.2	12.2	100.0
	Total	640	100.0	100.0	

The same frequency table in APA style.

Highest level of education completed

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No formal schooling	40	6.3	6.3	6.3
	Primary	64	10.0	10.0	16.3
	Secondary	158	24.7	24.7	40.9
	Higher secondary	148	23.1	23.1	64.1
	Graduate	152	23.8	23.8	87.8
	Postgraduate	78	12.2	12.2	100.0
	Total	640	100.0	100.0	

And in SPSS style.

Copy one table into Word

1. Choose **APA tables** at the top of Output.
2. Click **Copy table** under the table you want (or **Copy** at the top of a result to copy all of it).
3. Paste into Word or Google Docs with **Ctrl+V**. The table keeps its lines and formatting.

Excel, next to **Copy table**, saves that one table as a spreadsheet.

Copy the APA sentence

Click **Copy** in the **APA-style report** box and paste the sentence into your results section. Treat it as a draft:

- Rewrite long question wording into short concepts your reader knows ("feeling safe after dark").
- Put statistical symbols in italics: *t*, *F*, *p*, *M*, *SD*, *N*, *r*, *d*. Socius copies plain text, so italics are lost.
- Add the direction and meaning of the result in your own words.

Export the whole report

Click **Export report** in the Output tab, or choose **File > Export output report**, and pick a format:

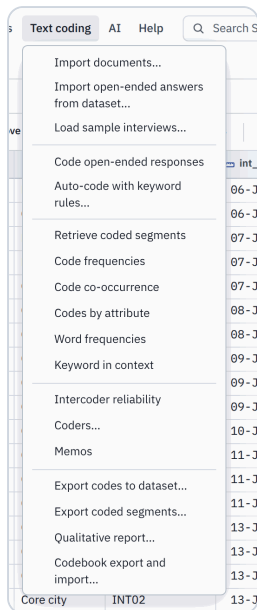
- **Word document (.docx)**: APA tables and charts, ready to edit. This is the one most people want.
- **Web page (.html)**: a standalone page you can open in any browser and print, or save as PDF from the print dialog.
- **Excel workbook (.xlsx)**: one sheet per table.
- **Plain text (.txt)**: tables as aligned text.

To tidy up before exporting, use the small arrows on each result to move it up or down, and the bin icon to delete it. After a delete, a message appears with an **Undo** button that brings the result back; **Edit > Undo** (**Ctrl+Z**) in the Output tab does the same. **Clear output** removes everything.

Coding open-ended answers

Surveys often end with an open question, such as "What is the biggest challenge facing your neighbourhood today?". *Coding* means reading each answer and tagging it with one or more themes, called **codes**. The list of codes, with a definition for each, is your **codebook**. Once answers are coded, you can count the themes, compare them between groups, and even test the differences with statistics.

Everything for coding is in the **Text coding** menu and the **Text coding** tab. The optional AI helpers for coding are in the **AI** menu (see AI in Text coding (page 56)).



The Text coding menu: import, code, analyse and export. The AI helpers are in the AI menu.

Start with the worked example

The quickest way to learn is the worked example. Click the **Text coding** tab. With the sample survey open, the start screen of Text coding shows a card called **Explore a worked example**.

Click it. Socius brings in the 630 answers to [q_challenge](#), adds a starter codebook of 15 codes grouped into 4 themes, and codes most answers automatically with keyword rules, so you have something to review. A note above the answers explains what happened and lists the next steps, with buttons such as **Review coded answers**, **Show answers not coded** and **Codes by attribute**; **Hide note** puts it away. Straight after loading, one click on **Undo coding** in the Text coding toolbar removes the whole example again.

Import answers from a survey question

For your own data, bring in the answers yourself:

1. Choose **Text coding > Import open-ended answers from dataset**.
2. Under **Open-ended question (string variable)**, pick the question, for example [q_challenge](#).

3. Under **Name responses by (optional)**, pick an ID such as `resp_id`.
4. Tick the characteristics you want to compare later, such as `gender`, `city`, `area` and `migrant`. These travel with each answer as *attributes*.
5. Click **Import 630 answers** (the button shows how many answers will come in).

Import sources
Interview transcripts, field notes and open-ended survey answers

Files

Paste text

Sample interviews

Open-ended answers in the dataset

Open-ended question (string variable)

What is the biggest challenge facing your neighbourhood today? (open-ended) [c

Name responses by (optional)

resp_id

Attributes (respondent characteristics to compare codes by)

☐ resp_id Respondent ID
☒ city City
☒ area Type of area
☐ interviewer Interviewer ...
☒ gender Gender of respon...

☐ educ Highest level of ed...
☐ employ Main employe...
☐ marital Marital status
☐ hh_income Total monthl...
☒ migrant Were you born i...

☐ trust1 Most people in thi...
☐ trust2 People around her...
☐ trust3 You have to be ver...
☐ trust3_r You have to be v...
☐ trust4 I trust the local m...

☐ trust5 I would feel safe ...
☐ civic_meet In the last 12...
☐ civic_vol In the last 12 m...
☐ civic_petition In the last ...
☐ civic_contact In the last ...

☐ civic_protest In the last ...
☐ belong How strongly do ...
☐ life_sat All things consid...
☐ health In general, how w...
☐ internet Do you use the i...

☐ discrim In the last 12 mo...
☐ vote Did you vote in the l...
☐ agegrp Age group

Case 1 Poor lighting on the approach road makes it scary after sunset.

Case 2 monsoon flooding, corruption

Case 3 housing, no jobs for youth

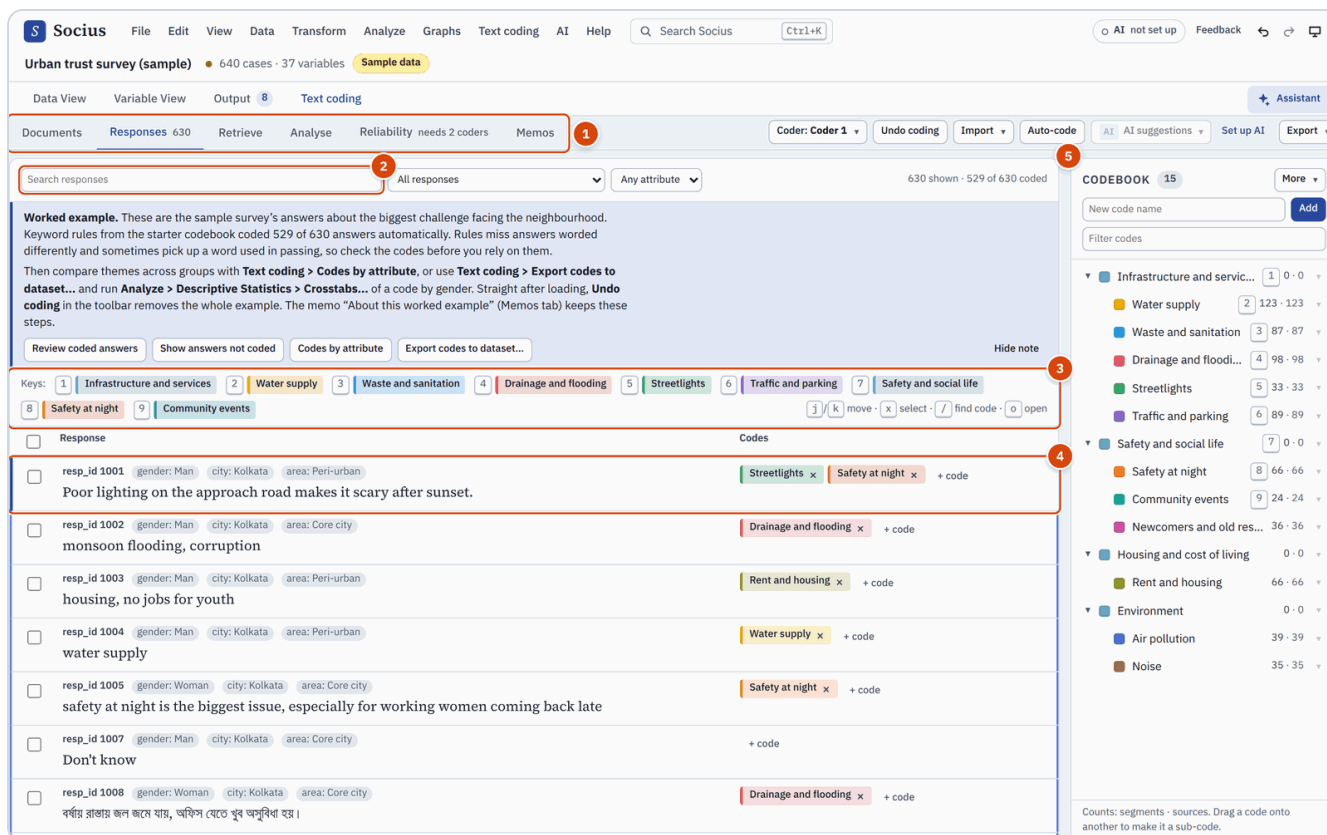
Case 4 water supply

Import 630 answers

Importing the answers to `q_challenge` with four attributes.

The Responses view

The answers appear in the **Responses** view of the Text coding tab.



The Responses view with the worked example loaded.

- Views.** Switch between Documents, Responses, Retrieve, Analyse, Reliability and Memos. Reliability says "needs 2 coders" until a second coder has coded. On a phone this row scrolls sideways; a fade and an arrow at the edge show that there is more.
- Search and filters.** Find answers with a word, show only answers not yet coded, or only one group (for example only women).
- Number keys.** The first nine codes get the keys 1 to 9.
- An answer** with its attributes and its codes. Click **+ code** to add a code with the mouse, or the small cross on a code to remove it.
- Codebook.** Your codes, grouped under themes, with how many passages and sources each has.

Build a codebook

To add a code, type its name in **New code name** at the top of the Codebook panel and click **Add**. Start with a handful of codes from reading 30 or 40 answers; you can add more as you go.

Good codes have clear definitions. Open a code's menu (the small arrow at the right of its row) and choose **Edit definition and rules**.

Edit code: Water supply
123 coded segments

Name
Water supply

Theme (parent code)
Infrastructure and services

Colour
#e9b200
Sample highlight

Definition
Not enough water, irregular or unpredictable supply, low pressure, dirty tap water, or dependence on tankers and cans.

Include when
Drinking and household water: timing, quantity, quality, cost of tankers.

Exclude when
Rain water and waterlogging (code Drainage and flooding).

Example
We depend on tankers for drinking water and they charge 800 rupees a trip.

Auto-coding rules (one per line)
English
water supply
water supply
water problem

Plain words match whole words, case-insensitive; * is a wildcard; several words make a phrase; /pattern/i is a regular expression. Run them from Auto-code.

Delete
Merge into...
Cancel
Save

Editing a code: its theme, definition, when to include and exclude it, an example and keyword rules.

- **Theme (parent code)** groups codes into a hierarchy, such as "Water supply" under "Infrastructure and services".
- **Definition, Include when, Exclude when** and **Example** keep your coding consistent over weeks, and they are exactly what a second coder needs.

Code with the number keys

Coding by keyboard is fast. Click in the list of answers, then:

Key	What it does
j or ↓	Next answer
k or ↑	Previous answer
1 to 9	Apply that code; press again to remove it
/	Find any code by name, or create a new one
x or Space	Select an answer (to code several at once)
o or Enter	Open the answer to code only part of it

An answer can have several codes. The counter at the top right shows how many answers you have coded so far.

Made a mistake? **Undo coding** in the toolbar takes back your last coding change. In the Text coding tab, **Edit > Undo** (**Ctrl+Z**) does the same, and **Edit > Redo** (**Ctrl+Y**) brings the change back. The Edit menu says what it will undo, for example "Undo code passage". Undo in Text coding only touches coding (codes, coded passages, memos and sources), never your data.

Auto-code with keyword rules

For clear-cut themes, rules save time. Choose **Text coding** > **Auto-code with keyword rules** (or the **Auto-code** button).

Auto-code with keyword rules
Find passages by keywords or patterns, check them in context, then code them in one step.

Infrastructure and services 31
Water supply 17
Waste and sanitation 27
Drainage and flooding 13
Streetlights 15
Traffic and parking 20
Safety and social life 13
Safety at night 14
Community events 17
Newcomers and old r... 17
Housing and cost of living
Rent and housing
Environment

Rules for "Safety at night" (one per line)

English
safe
unsafe
safety
eve teasing
chain snatching
police
safe road

Plain words match whole words, ignoring case; * matches any ending (migr* finds migrant, migration); several words form a phrase; /pattern/i is a regular expression; lines starting with # are notes.

What gets coded: the whole response or document
Search in: all sources

11 codes with rules will be run. New segments are marked as auto-coded and belong to Coder 1. Passages already coded with the same code are skipped.

Close Preview matches

Keyword rules for the code "Safety at night".

1. Pick a code on the left.
2. Type its rules on the right, one per line. A plain word matches that whole word, ignoring capitals. * matches any ending (flood* finds flood, floods and flooding). Lines starting with # are notes to yourself.
3. Choose **What gets coded** (the whole answer, or only the sentence with the match) and **Search in**.
4. Click **Preview matches**, untick any wrong hits, then click the button to code them.

ALWAYS CHECK AUTOMATIC CODES

Keyword rules miss answers that describe a theme in other words or in another language, and they sometimes catch a word used in passing. Read a sample of the auto-coded answers, and use the filter **Not coded yet** to find the ones the rules missed.

Turn codes into variables and crosstab them

This is where qualitative and quantitative work meet. Each code can become a 0/1 variable in your dataset (1 = the person mentioned it, 0 = did not).

1. Choose **Text coding** > **Export codes to dataset**.
2. Tick the codes you want. Each gets a variable name starting with c_, such as c_safety_at_night.
3. Click **Add ... variables**.

Export codes to dataset

One 0/1 variable per code, so you can run crosstabs or regressions on what people wrote

Codes

Select none

Code	New variable	Mentioned
☒ Infrastructure and services theme: 1 if any of its sub-codes applies	c_infrastructure_and	367 of 630
☒ Water supply	c_water_supply	123 of 630
☒ Waste and sanitation	c_waste_and_sanitation	87 of 630
☒ Drainage and flooding	c_drainage_and_flooding	98 of 630
☒ Streetlights	c_streetlights	33 of 630
☒ Traffic and parking	c_traffic_and_parking	89 of 630
☒ Safety and social life theme: 1 if any of its sub-codes applies	c_safety_and_social_life	126 of 630
☒ Safety at night	c_safety_at_night	66 of 630
☒ Community events	c_community_events	24 of 630
☒ Newcomers and old residents	c_newcomers_and_old	36 of 630
☒ Housing and cost of living theme: 1 if any of its sub-codes applies	c_housing_and_cost_of	66 of 630
☒ Rent and housing	c_rent_and_housing	66 of 630
☒ Environment theme: 1 if any of its sub-codes applies	c_environment	71 of 630
☒ Air pollution	c_air_pollution	39 of 630
☒ Noise	c_noise	35 of 630
☒ Also add a count variable (number of these codes mentioned)		

Values: 1 = Mentioned, 0 = Not mentioned (value labels included), system-missing for cases with no answer. Measurement level nominal.

Cancel
Add 16 variables

Exporting codes as variables. The last column shows how many answers mention each code.

The new variables are a change to your data, so you take them back from Data View with **Edit > Undo**, not from Text coding.

Now run **Analyze > Descriptive Statistics > Crosstabs** with **gender** in **Row(s)** and **c_safety_at_night** in **Column(s)**:

Table 25

*gender * c_safety_at_night Crosstabulation*

Gender of respondent by Safety at night

			c_safety_at_night		
			Not mentioned	Mentioned	Total
gender	Man	Count	308	13	321
		% within gender	96.0%	4.0%	100.0%
	Woman	Count	240	50	290
		% within gender	82.8%	17.2%	100.0%
	Other / prefer to self-describe	Count	16	3	19
		% within gender	84.2%	15.8%	100.0%
Total		Count	564	66	630
		% within gender	89.5%	10.5%	100.0%

Copy table Excel

Table 26

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	28.859 ^a	2	< .001
Likelihood Ratio	30.600	2	< .001
Linear-by-Linear Association	25.530	1	< .001
N of Valid Cases	630		

Note. a. 1 cells (16.7%) have expected count less than 5. The minimum expected count is 1.99.

Copy table Excel

Women mention safety at night far more often than men.

17.2% of women mention safety at night as the biggest challenge, against 4.0% of men, $\chi^2(2, N = 630) = 28.86, p < .001$. The answer to an open question has become a testable finding.

To see how often each code was used, choose **Text coding > Code frequencies**. A theme shows the total of its sub-codes, marked "theme total". To compare codes between groups without leaving Text coding, open the **Analyse** view and choose **Codes by attribute**:

Code frequenciesCo-occurrence**Codes by attribute**Word frequenciesKeyword in context

AttributegenderSend to Output

Code	Man (n = 321)		Woman (n = 290)		Other / prefer to self-describe (n = 19)	
	Count	Col %	Count	Col %	Count	Col %
Infrastructure and services with sub-codes	197	61.4%	163	56.2%	7	36.8%
Water supply	71	22.1%	50	17.2%	2	10.5%
Waste and sanitation	46	14.3%	38	13.1%	3	15.8%
Drainage and flooding	47	14.6%	49	16.9%	2	10.5%
Streetlights	14	4.4%	19	6.6%	0	0.0%
Traffic and parking	52	16.2%	34	11.7%	3	15.8%
Safety and social life with sub-codes	47	14.6%	73	25.2%	6	31.6%
Safety at night	13	4.0%	50	17.2%	3	15.8%
Community events	14	4.4%	9	3.1%	1	5.3%
Newcomers and old residents	20	6.2%	14	4.8%	2	10.5%
Housing and cost of living with sub-codes	40	12.5%	23	7.9%	3	15.8%
Rent and housing	40	12.5%	23	7.9%	3	15.8%
Environment with sub-codes	41	12.8%	27	9.3%	3	15.8%
Air pollution	25	7.8%	13	4.5%	1	5.3%
Noise	18	5.6%	15	5.2%	2	10.5%

Column % = share of the responses in that group coded with the code. To test differences, use Export codes to dataset and run Crosstabs.

Codes by attribute: how often each code appears among men, women and people of other genders.

CHAPTER 11

Coding interviews

Interview transcripts are coded in the **Documents** view. You highlight a passage and attach one or more codes to it.

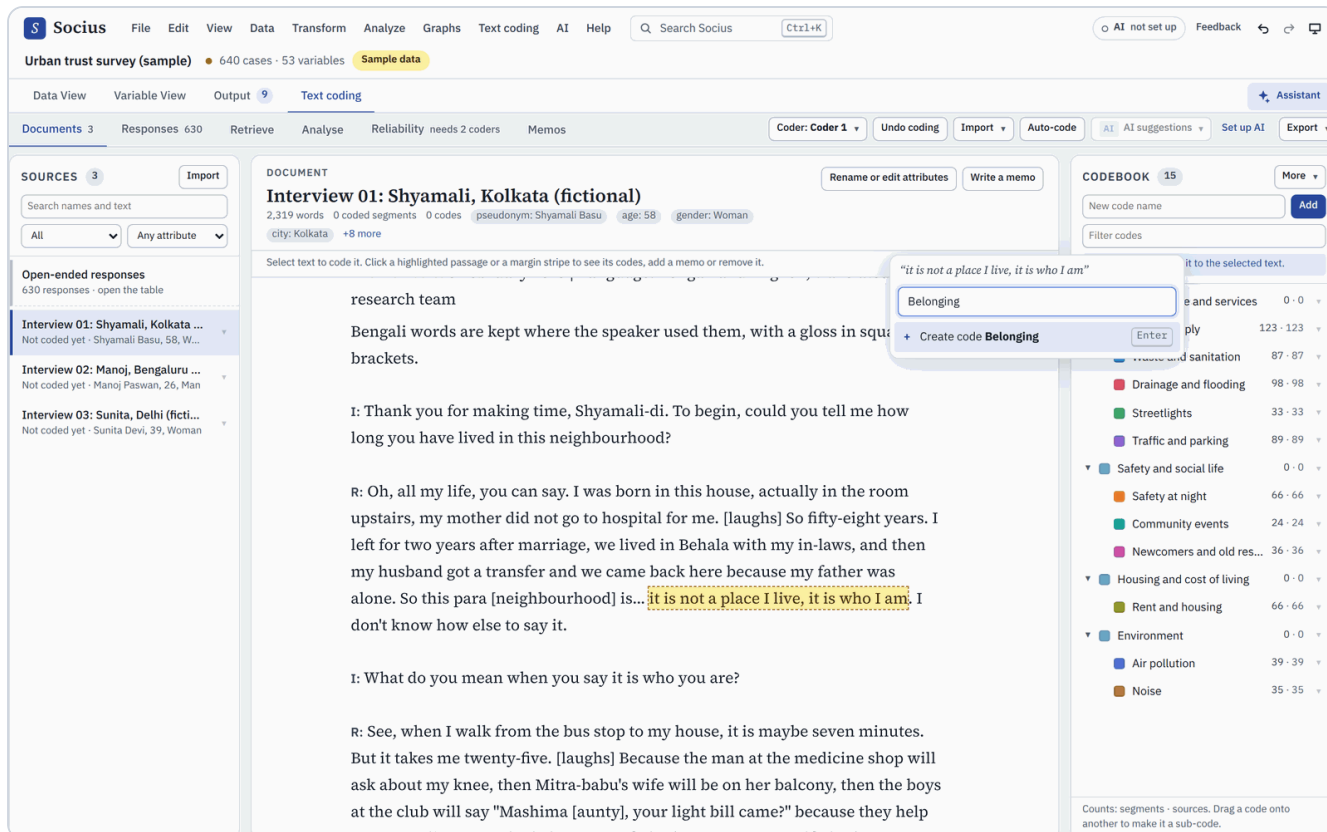
Import a transcript

Choose **Text coding > Import documents**. Drop your files on the dialog, or click to choose them.

- **Word (.docx)** and **plain text (.txt, .md)** files: one document per file.
- **Paste text** lets you paste a transcript directly.
- To practise, choose **Text coding > Load sample interviews** and click **Load 3 interviews**. It adds three fictional interviews from Kolkata, Bengaluru and Delhi. The same command is under **Import** in the Text coding toolbar.

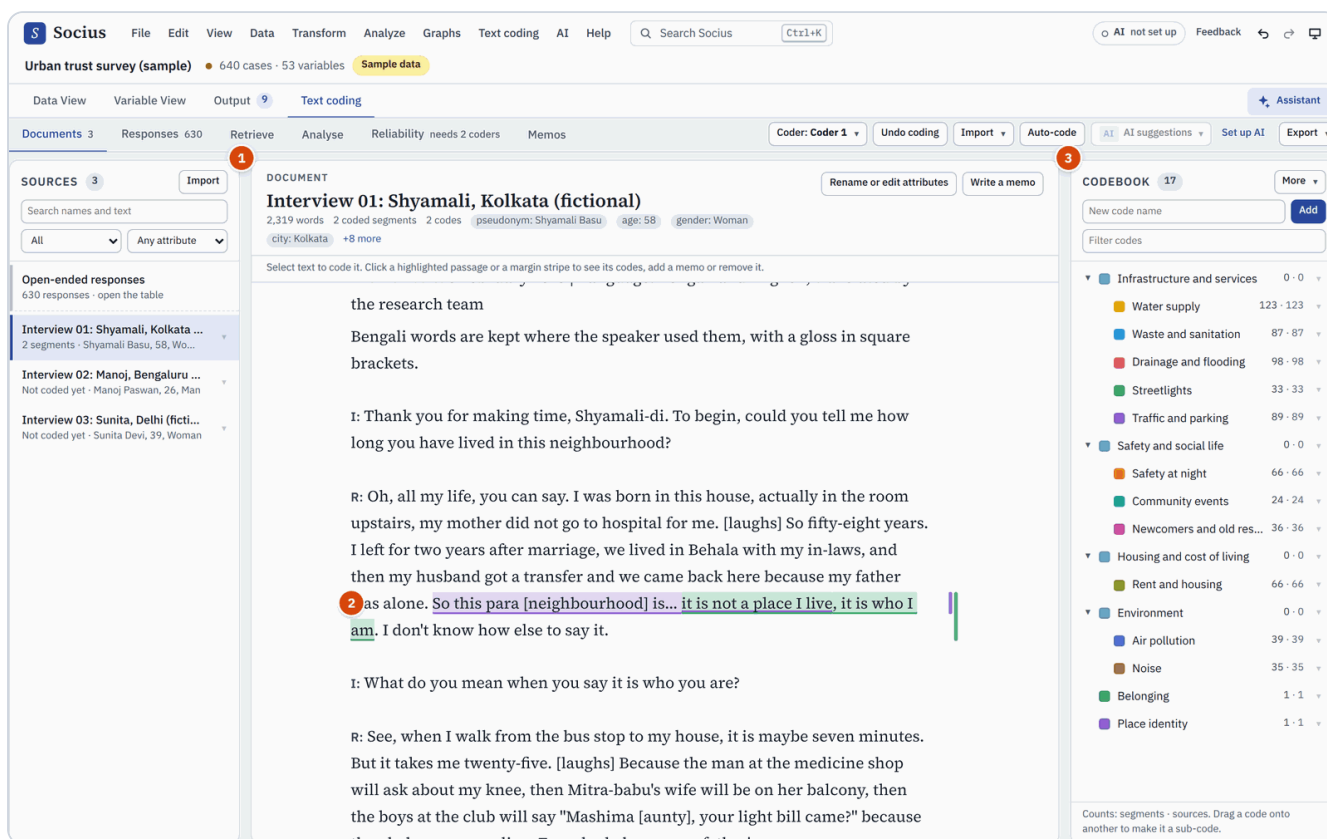
Highlight and code a passage

1. Open the **Documents** view and click an interview in the **Sources** panel on the left.
2. Select a passage with the mouse, as you would to copy it.
3. A small box appears. Type the name of a code. Press to apply an existing code, or choose **Create code** to add a new one on the spot.



Selecting a passage and creating the code "Belonging".

Coded passages are highlighted in the code's colour, with a stripe in the margin. A passage can have several codes, and codes can overlap.



A coded interview: the Sources (1), coded passages (2) and the Codebook (3).

Write memos

Memos are your analytic notes. There are two kinds:

- **A memo on a passage.** Click a highlighted passage. A panel shows its codes, with a box for a memo on each.
- **A memo on the project or a code.** Open the **Memos** view and click **New memo**, or choose **Write a memo on this code** from a code's menu.

The screenshot displays the Socius software interface in the 'Text coding' view. The main document is titled 'Interview 01: Shyamali, Kolkata (fictional)'. A highlighted passage is selected, and a panel shows two codes: 'Place identity' and 'Belonging'. A memo box is visible for the 'Place identity' code, containing the text: 'Identify fused with place: she speaks of the para as part of who she is.' The interface includes a top menu bar, a left sidebar with 'SOURCES' and 'Open-ended responses', and a right sidebar with a 'CODEBOOK'.

Clicking a coded passage shows its codes and a memo box for each.

Retrieve quotes

The **Retrieve** view collects every passage with a given code, across all your documents and answers, with the speaker's attributes. Pick a code at the top.

Safety at night

☒ Include sub-codes

Any attribute

Safety and social life > Safety at night

Copy quotes

Export CSV

Export Excel

Feeling unsafe after dark, harassment of women and girls, thefts, drinking in public, little police presence.

66 segments from 66 sources

Poor lighting on the approach road makes it scary after sunset.

resp_id 1001

gender: Man

city: Kolkata

area: Peri-urban

Coder 1 · Auto-coded (rule)

Show in context

Copy

safety at night is the biggest issue, especially for working women coming back late

resp_id 1005

gender: Woman

city: Kolkata

area: Core city

Coder 1 · Auto-coded (rule)

Show in context

Copy

The lane to the station is dark and lonely, I always call someone while walking. I have started taking an auto even for a short distance. Garbage is not collected regularly and it piles up at the corner of the road. Please do something about it.

resp_id 1016

gender: Woman

city: Kolkata

area: Core city

Coder 1 · Auto-coded (rule)

Show in context

Copy

Safety at night is the biggest issue, especially for working women coming back late. The only open ground was taken for a parking lot.

resp_id 1051

gender: Man

city: Kolkata

area: Core city

Coder 1 · Auto-coded (rule)

Show in context

Copy

According to me, the lane to the station is dark and lonely, I always call someone while walking.

All passages coded "Safety at night".

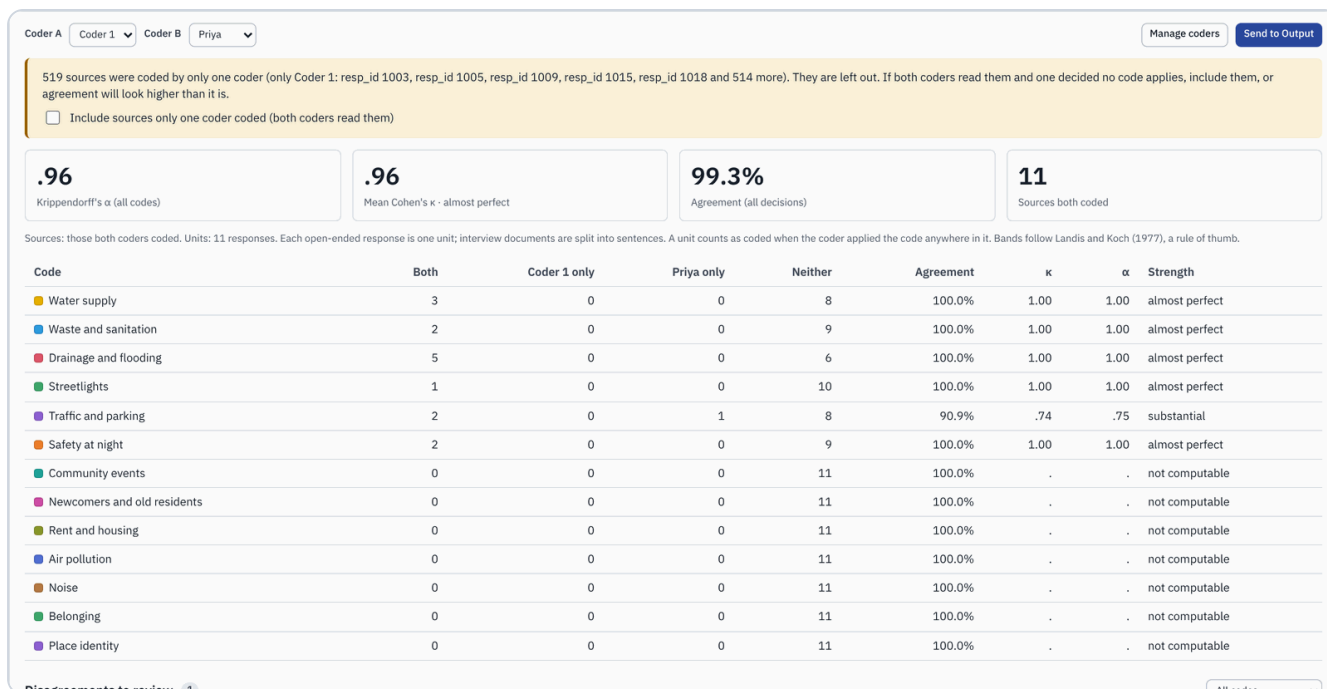
- **Copy quotes** copies them with their sources, ready for your write-up.
- **Show in context** jumps to the passage in its document.
- **Export CSV** and **Export Excel** save them as a table.

For a report of the whole codebook with counts and example quotes, choose **Text coding > Qualitative report**.

Intercoder reliability: do two coders agree?

To show that your coding is trustworthy, a second person codes the same material independently, and you measure how often you agree.

1. Agree on the codebook with written definitions first.
2. Choose **Text coding > Coders** (or **Coders** under the **Coder** button in the toolbar). Add the second coder, for example "Priya", and click **Code as Priya**.
3. Priya codes the same answers or documents. While she codes, she does not see the first coder's codes, so her coding stays independent.
4. If you work on different computers, one person saves the project (**File > Save project**) and sends the `.socius.json` file to the other, who opens it, codes, and saves it again.
5. Open the **Reliability** view and choose **Coder A** and **Coder B**.



Inter-coder reliability between Coder 1 and Priya.

How to read it.

- **Agreement** is the share of decisions on which both coders agree. It looks high even by chance, because both coders leave most codes off most answers.
- **Cohen's κ (kappa)** and **Krippendorff's α (alpha)** correct for agreement by chance. For published work, .70 to .80 or higher is usually expected. The **Strength** column gives a verbal label ("substantial", "almost perfect").
- **Disagreements to review**, further down, lists the answers where you differ. Discuss them, sharpen the code definitions, and code a fresh sample if needed.

CHAPTER 12

Getting help from AI

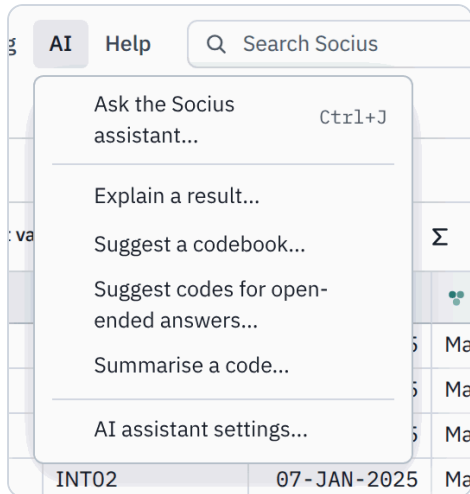
Socius can ask an AI model to help you. Once it is set up, AI can:

- **answer your questions** about your data and methods, and run the analysis for you (the Socius assistant);
- **explain a result** in plain language, with what to watch out for and how to report it;
- **help with text coding**: suggest a codebook, suggest codes for open-ended answers and summarise a code.

AI help is optional and free. It is switched off until you set it up, and everything else in Socius works without it. Setting it up takes about two minutes.

Set it up

1. Choose **AI > AI assistant settings**. This is the one place where AI help is set up. The **AI** chip at the top right and every **Set up AI** button (next to an AI feature that is greyed out) open the same settings.
2. Under **Where should the AI run?**, choose one of the free options below.
3. Follow the steps for that option, then click **Test connection**.



The AI menu. Every AI feature starts here, and the last item opens the settings.

Option 1: Google Gemini with your own free key (recommended)

Google Gemini is fast and gives good answers. You need a free "API key" from Google: a long password that lets Socius use Gemini on your behalf. You only need a Google account (for example a Gmail address).

1. In the settings, choose **Google Gemini** (1).
2. Click the link **Google AI Studio** (2), or go to aistudio.google.com/apikey. Sign in with your Google account.
3. Click **Create API key**. Accept the terms if Google asks.
4. Copy the key with the copy button next to it in AI Studio. New keys start with "AQ." and are long, for example `AQ.Ab8R...` followed by about 50 more characters. (Older keys start with "AIza". Google is retiring those, so if one stops working, create a new key.)
5. Back in Socius, paste it into **API key** (3). Socius removes stray spaces, line breaks and quotes for you. If the key looks too short or has an unusual start, an orange note under the box says so; you can still test it.
6. Decide whether Socius should **Remember this key on this computer** (4). See the box below.
7. Leave **Model** (5) on **Automatic: Flash-Lite (fastest, most free requests)**. Socius then picks the newest Flash-Lite model your key can use, and moves on to another one if Google retires it.
8. Click **Test connection** (6).

AI assistant
×

Optional help: ask the Socius assistant, explain a result in plain language, suggest a codebook, suggest codes for open-ended answers, summarise a code. Nothing is sent until you click a button that asks the AI.

WHERE SHOULD THE AI RUN?

☐ **On this computer** Free, private

Good: Nothing leaves your computer. Works offline after a one-time download.
But: Downloads 1 to 2 GB, needs a recent Chrome or Edge, and is slower and less accurate than online models.

☒ **Google Gemini** Free key

Good: Good quality and fast. The key is free with a Google account.
But: Excerpts go to Google, which may use free-tier data to improve its products.

☐ **Other service** Advanced

Good: Groq, OpenRouter, or a model on your own computer (Ollama, LM Studio).
But: Needs a key or a local setup. Check each service's terms.

Get a free key in 1 minute

1. Open [Google AI Studio](#) and sign in with a Google account.
2. Click **Create API key** (accept the terms if asked).
3. Copy the key with its copy button (new keys start with AQ .) and paste it below. It stays in this browser.

API key 3

Show

☐ **Remember this key on this computer** 4

Off: the key is forgotten when you close this tab. Anyone using this browser profile, and other sites hosted on hackhead95.github.io, could read a remembered key. Leave it off on shared computers.

Forget key Removes the key from this browser. Use this on a shared computer.

The excerpts you send go to Google. On the free tier, Google may use what you send to improve its products, and human reviewers may read it. Do not send identifiable or confidential data: anonymise excerpts first (names, places, employers, rare details), and check that your participants' consent and your ethics approval allow sharing data with an outside service. Consider the on-device option for sensitive material.

Test connection 6

AI suggestions are a starting point for your own reading, not findings. Check every suggested code and quote against the data, and report in your methods section that AI assistance was used and how.

Saved in this browser only, never in project files.

Done

Setting up Google Gemini: choose it (1), get a key from Google AI Studio (2), paste it (3), decide whether to remember it (4), leave Model on Automatic (5) and click Test connection (6).

WHERE YOUR KEY IS KEPT

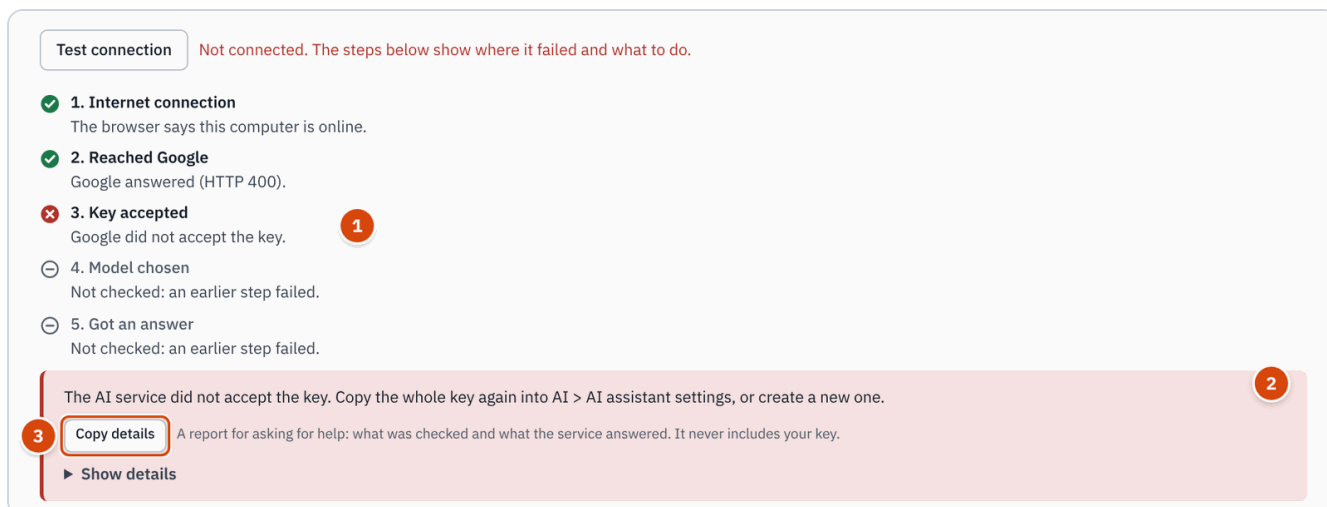
The key never goes into project files or exports. By default, **Remember this key on this computer** is off: Socius forgets the key when you close the tab, and you paste it again next time. Tick the box to keep it in this browser until you click **Forget key**. Anyone who uses the same browser profile, and other websites published on the same github.io address as Socius, could read a remembered key, so leave the box off on shared or public computers.

Which model? Automatic: Flash-Lite answers fastest and allows the most free requests per day, so it is the best choice for almost everyone. **Automatic: Flash** writes somewhat better answers but starts more slowly, and Google's free tier allows only about 5 requests a minute with it. The assistant's working steps and the coding suggestions always use Flash-Lite. **A model I type...** is for people who need one particular model.

Test connection, step by step

Test connection sends a one-word test message and checks the connection one step at a time: **1. Internet connection**, **2. Reached Google**, **3. Key accepted**, **4. Model chosen** and **5. Got an answer**. Each step gets a tick when it passes. While the test runs, the line next to the button says what it is waiting for and for how long. **Stop** cancels it.

If a step fails, it gets a cross (1), the steps after it are marked "Not checked", and a red box (2) explains what went wrong and what to do.



A refused key: steps 1 and 2 passed, step 3 failed (1). The red box (2) says what to do; Copy details (3) copies a report for a help request.

Copy details (3) copies a report of what was checked and what Google answered, to paste into a help request (see Giving feedback (page 62)). The report never contains your key. **Show details** shows the same report on screen. The most common problems and their fixes are listed in Troubleshooting and FAQ (page 59) under "Test connection fails".

Option 2: On this computer (private)

A small AI model runs inside your browser, on your computer's graphics chip. Nothing leaves your computer, and after a one-time download of 1 to 2 GB it even works offline. This is the right choice for confidential interviews.

1. In the settings, choose **On this computer**.
2. Socius checks whether your browser can run the model and says so at the top. It needs a recent **Chrome or Edge** on a desktop or laptop, with graphics acceleration turned on. If something is missing, the message says what.
3. Pick a model: **Small and fast** (a download of about 1 GB) or **Better quality** (about 1.8 GB). On a computer with little memory, Socius recommends Small and fast.
4. Click **Download model**. A bar shows the progress. If the download fails, **Try the download again** keeps the parts that already arrived.
5. Click **Test connection**.

Before a download starts, Socius checks that the browser has room for the model and still has space left for autosave. If not, an orange box says **Not enough browser storage for this model** and suggests what to do instead: delete a model you downloaded earlier, choose Small and fast, use Google Gemini or Ollama, or free disk space. **Browser storage (downloaded models)**, at the bottom of this section, shows how much room each downloaded model takes, with a **Delete** button for each (see What Socius keeps in your browser (page 57)).

The on-device model is slower and less accurate than Gemini. It is fine for explaining a result or suggesting codes, but it is noticeably weaker as the Socius assistant.

Option 3: Ollama or LM Studio on your computer (for experienced users)

If you already use [Ollama](#) or LM Studio, programs that run AI models on your own computer, Socius can use them. Nothing leaves your computer. This takes a few more steps than the other options, but Socius guides you through them.

1. In the settings, choose **Other service**.
2. Under **Service**, choose **Ollama on this computer** or **LM Studio on this computer**. Socius fills in the address (for Ollama <http://localhost:11434/v1>) and a model.
3. Follow the steps in **Set up Ollama** (1) (or **Set up LM Studio**): install the program, download a model once (for Ollama, the box shows the command, such as `ollama pull llama3.2`, with a **Copy** button), and let the program accept this website.
4. Click **Test connection** (2).

The test checks five things: **Is the program running?** (3), **Does it allow this website?** (4), **Browser permission for this computer**, **Is the model installed?** and **Did it answer?**. When a step fails, it shows the fix for your computer.

▼ Set up Ollama 1

1. Install Ollama from ollama.com and open it.
2. Download the model once (about 2 GB for llama3.2), in a terminal:
`ollama pull llama3.2` **Copy**
3. Let Ollama accept this website (<https://hackhead95.github.io>). Test connection shows the exact steps for your computer if this is still needed.
4. In Chrome or Edge, if the browser asks whether this site may connect to apps or devices on your computer, click **Allow**.
5. Click **Test connection**.

Test connection 2 **Copy details**

✓ Is the program running? 3
Ollama is running at <http://localhost:11434>.

✗ Does it allow this website? 4
Ollama is running but refused this website (<https://hackhead95.github.io>). Ollama only accepts requests from pages on this computer unless you add this website's address to the OLLAMA_ORIGINS setting and restart Ollama.

Tell Ollama to accept this website, <https://hackhead95.github.io>, by setting `OLLAMA_ORIGINS` and restarting Ollama.

Windows **macOS** **Linux (this computer)** 5

1. Quit Ollama: right-click the Ollama icon near the clock (in the taskbar corner; click ^ if you cannot see it) and choose Quit Ollama.
2. Open Command Prompt or PowerShell (press the Windows key, type cmd, press Enter) and run this command:
`setx OLLAMA_ORIGINS "https://hackhead95.github.io"` **Copy**
3. Start Ollama again from the Start menu.
4. Or, without a command: Start > search "environment variables" > "Edit environment variables for your account" > New. Name: OLLAMA_ORIGINS, value (exactly, no spaces):
`https://hackhead95.github.io` **Copy**
5. If you start Ollama yourself in PowerShell instead, start it like this:
`$env:OLLAMA_ORIGINS="https://hackhead95.github.io"; ollama serve` **Copy**

Type the address exactly, with no path and no spaces. Only add websites you trust: any website listed there can use your Ollama. If OLLAMA_ORIGINS already lists other addresses, separate them with commas and no spaces.

✓ Browser permission for this computer
Allowed: this website may connect to programs on this computer.

☐ **Is the model installed?**
Needs the program to allow this website first.

☐ **Did it answer?**
Needs the program to allow this website first.

Ollama is running (3) but does not accept the Socius website yet (4). The buttons (5) show the exact steps for Windows, macOS or Linux.

Letting Ollama accept Socius. Ollama only answers pages on your own computer, so at first it refuses the Socius website. The fix is a setting called `OLLAMA_ORIGINS` that lists the website's address: the start of the address in your browser's address bar, without the rest of the path and without spaces. For the Socius website that is <https://hackhead95.github.io>. Click **Windows**, **macOS** or **Linux** (5): Socius shows the steps with the right address filled in, and each command has a **Copy** button. In short:

- **Windows:** quit Ollama (right-click its icon near the clock and choose **Quit Ollama**), run the `setx OLLAMA_ORIGINS ...` command in Command Prompt, then start Ollama again from the Start menu. Instead of the command, you can add the setting under **Edit environment variables for your account**.
- **macOS:** run the `launchctl setenv OLLAMA_ORIGINS ...` command in Terminal, then quit and reopen Ollama. This setting is lost when the Mac restarts, so run the command again after a restart.
- **Linux:** add the `OLLAMA_ORIGINS` line to the Ollama service with `sudo systemctl edit ollama.service`, then restart the service.

Only list websites you trust: any website in that list can use your Ollama. For **LM Studio**, open its Developer tab, start the server and turn on **Enable CORS** in its settings instead.

The browser's permission. Chrome and Edge ask before a website may connect to programs on your computer. When the question appears, click **Allow**. If you clicked Block by mistake, click the icon at the left of the address bar, choose **Site settings**, and set **Apps on device** (called **Local network access** in some versions) to **Allow**, then reload the page. Safari does not let websites reach programs on your computer at all: use Chrome, Edge or Firefox for Ollama and LM Studio.

OTHER SERVICES

Under **Other service** you can also connect Groq or OpenRouter with their own keys. If you open Socius inside Claude, a **Claude** option may also appear, which needs no set-up.

"AI is ready. Try it"

When the test works, the settings say "Connected" (1), every step has a tick (2), and a green panel appears (3).

Test connection Connected to gemini-3.6-flash-lite. The AI answered: "OK" 1

- ✓ 1. Internet connection
The browser says this computer is online.
- ✓ 2. Reached Google
Google answered.
- ✓ 3. Key accepted
Accepted. Google lists 2 models for this key, 2 suitable for text.
- ✓ 4. Model chosen
gemini-3.6-flash-lite (automatic, best of 2 suitable models)
- ✓ 5. Got an answer
gemini-3.6-flash-lite answered: "OK"

► Details

AI is ready. Try it: 3

Ask the Socius assistant Explain a result Suggest a codebook Suggest codes for open-ended answers Summarise a code

Each one shows what will be sent before anything goes to the AI.

After a successful Test connection (1), all five steps are ticked (2) and the panel "AI is ready. Try it" (3) starts any AI feature with one click.

Click one of the buttons to try a feature straight away, or click **Done**. The **AI** chip in the top bar now shows a green dot and the service, for example "AI Gemini". **Details**, under the steps, holds the same report as **Copy details**.

Before you send anything

PROTECT YOUR PARTICIPANTS

With Google Gemini (or another online service), what you send leaves your computer and goes to that company. On free tiers, the provider may use it to improve its products, and people may read it. Before you use online AI help on real data:

- **Know what is sent.** Explain a result sends the result's tables and summary, never individual answers. The assistant sends variable information, summary statistics and analysis results; individual cases only if you switch that on. Text coding features send the excerpts you choose.
- **Anonymise first.** Remove names, places, employers and other details that could identify someone from texts before you send them.
- **Check consent and ethics approval.** Make sure your participants' consent and your ethics approval allow sharing data with an outside service.
- **Use On this computer for confidential material,** where nothing leaves your computer.

Every AI feature says what it will send, and to whom, before you click. Nothing is sent until you do.

Where to find AI in Socius

The **AI menu** in the menu bar lists everything AI can do:

Menu item	What it does
Ask the Socius assistant	Opens the assistant, where you can ask questions in your own words (Ctrl+J).
Explain a result	Explains one result from Output in plain language. With several results, you choose which one.
Suggest a codebook	Reads a sample of your interviews or open-ended answers and proposes codes with definitions and example quotes.
Suggest codes for open-ended answers	Applies your codebook to survey answers and suggests codes for each answer, which you accept or reject.
Summarise a code	Drafts a short summary of the passages coded with one code.
AI assistant settings	Set-up, as described above.

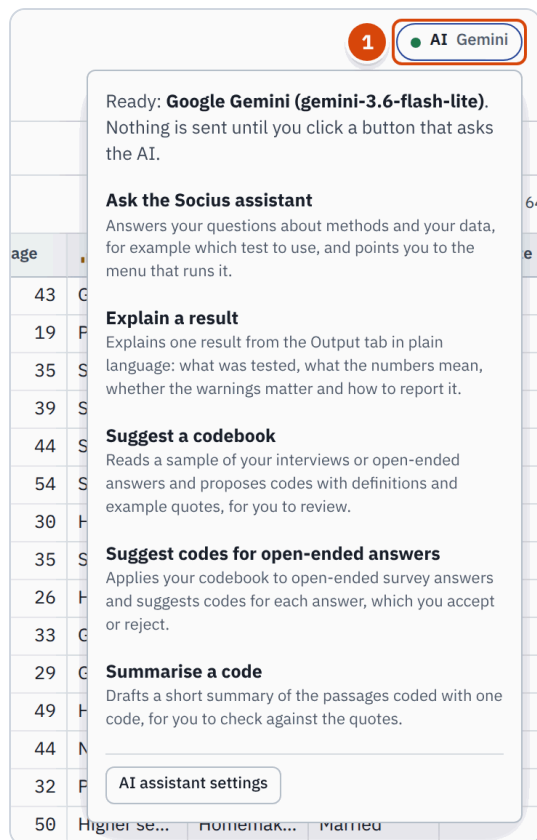
If something is missing, the menu tells you what to do first. For example, **Explain a result** with no results yet says **Run an analysis first** and offers a button **Open Crosstabs**. If AI is not set up, any item opens the settings and explains what that feature will do.

The **AI chip** at the top right shows the status of AI help. Click it to see the same features, each with a one-line description. It has four states:

- **not set up:** no AI option has been chosen yet.

- **Gemini, not tested** (a dashed dot): set up, but nothing has been tried yet. A new key or model starts here again.
- **Gemini, not connected** (a red dot): the last Test connection or AI request failed. Click the chip to see why.
- **Gemini** (a green dot): ready. A successful Test connection or AI answer puts the chip here.

Other services show their own name instead of "Gemini", such as "On device".



The AI chip (1) with a green dot means AI help is ready. Click it for a list of the AI features.

You also find AI in three other places: the **Assistant** button at the right end of the row of tabs, the **Explain with AI** button on every result in Output, and **AI suggestions** in the Text coding toolbar. All three are described below.

Explain a result

Every result in Output has an **Explain with AI** button. It asks the AI to explain the result as a patient statistics tutor would: what was tested, what the numbers mean, whether the warnings matter and how to report it.

Try it with the crosstab from Crosstabs with chi-square (page 20) ([gender](#) by [trust5](#)):

1. Run the crosstab, or scroll to it in Output.
2. Click **Explain with AI** at the top right of the result, next to **Copy** (see the picture below).
3. A panel opens under the result. It says what will be sent and to whom (1 in the second picture). Click **What will be sent** (2) to read the exact text. The tables, Socius's summary, the APA sentence and the warnings are sent; individual answers are not.
4. Click **Explain** (3).

Crosstabs
Urban trust survey (sample) · 24 Sep 2026, 18:42 · N = 630; 10 cases excluded for missing values.

1

Copy
AI Explain with AI
↑
↓

Explain with AI (1) is next to Copy on every result.

AI Explain this result with AI

The AI explains what was tested, what the numbers mean for your research question, whether the warnings matter, how to report the result, and what not to conclude from it.

1 When you click Explain, **the tables, Socius's summary, the APA sentence and the warnings of this result (no individual answers)** will be sent to **Google Gemini (gemini-3.6-flash-lite)**. Anonymise names and places first if the data is confidential. [AI assistant settings](#)

2 ▼ What will be sent (4 KB)

You are a patient statistics tutor helping a sociology student understand one result from Socius, a data analysis tool that works like SPSS.

Explain the result below in plain English for someone who has taken one introductory statistics course. Use only the numbers given here and never invent values. If something you would need is missing, say so.

Write five short sections, each starting with its heading on its own line:

What was tested

What the numbers mean

Assumptions and warnings

How to report it

Cautions

In "What the numbers mean", say which numbers matter most and what they show about the people or groups studied, quoting the actual values.

In "Assumptions and warnings", say whether the warnings or assumptions matter for the conclusion (for example small expected counts, unequal variances, small groups).

3 Explain Cancel

Discuss with the assistant

Before anything is sent: who receives what (1), the exact text (2) and the Explain button (3).

While the AI works, a line shows what it is doing and for how long, such as "Choosing model", "Waiting for Google", "Thinking" or "Writing". The answer appears bit by bit as it is written. If something goes wrong, a red message says why and what to do; the **Details** link under it shows a report, and **Copy details** copies it for a help request.

The explanation appears under five headings: **What was tested**, **What the numbers mean**, **Assumptions and warnings**, **How to report it** and **Cautions**.

AI

AI explanation

AI-generated: check against the tables

1

×

What was tested

Whether men, women and people of other genders differ in how safe they feel walking alone in their neighbourhood after dark. The chi-square test asks whether the answers could be independent of gender.

What the numbers mean

- Men were more likely to feel safe: **44.8%** agreed or strongly agreed, against **29.3%** of women.
- Women were more likely to strongly disagree (14.5% against 5.3% of men).
- $\chi^2(8, N = 630) = 35.02, p < .001$: a difference this large is very unlikely if gender and feeling safe were unrelated.
- Cramér's $V = .17$: the relationship is real but **weak**.

Assumptions and warnings

3 cells (20.0%) have an expected count below 5. They are in the small "Other / prefer to self-describe" group (19 people). That is at the usual 20% limit, so the test is acceptable, but do not read much into the percentages for that group.

How to report it

A chi-square test of independence showed a significant association between gender and feeling safe walking alone after dark, $\chi^2(8, N = 630) = 35.02, p < .001$, Cramér's $V = .17$. Men were more likely than women to agree that they felt safe (44.8% vs. 29.3%).

Cautions

- This is an association, not a cause. Experiences of harassment, street lighting or daily routines may lie behind it.
- The answers are self-reported, from one point in time.

Written by Google Gemini (gemini-3.6-flash-lite). It can be wrong: check every number against the tables above.

Copy

Add to output

Explain again

Discuss with the assistant

An explanation of the crosstab, labelled as AI-generated (1). Add to output (2) keeps it with the result; Discuss with the assistant (3) lets you ask follow-up questions. Example answer; your AI's wording will differ.

- **Add to output** (2) attaches the explanation to the result as a note marked **AI-GENERATED**. It is exported with your report like any note.
- **Copy** copies the text; **Explain again** asks for a new version.
- **Discuss with the assistant** (3) opens the Socius assistant with this result attached, so you can ask follow-up questions such as "What does Cramér's V mean here?".

READ IT CRITICALLY

The AI can be wrong, even when it sounds sure. Check every number against the tables above it: here $\chi^2(8, N = 630) = 35.02$ and $V = .17$ match the Chi-Square Tests and Symmetric Measures tables. Watch for claims the tables cannot support, such as causes ("gender causes fear"). Use the explanation to understand your result, then write the results section in your own words.

The Socius assistant

The assistant is a chat panel where you ask questions in your own words, as you would ask a helpful methods tutor. It can look at your dataset, run analyses on it, read your results, prepare recodes and scales, and answer "how do I" questions from this guide.

Open it

- Click the **Assistant** button at the right end of the row of tabs (number 9 in the tour (page 4)). On a phone it shows only its sparkle icon.
- Or press `Ctrl+J` (`Cmd+J` on a Mac). Press it again to close the panel.
- Or choose **AI > Ask the Socius assistant**, or type a question in Search and choose **Ask the assistant**.

The panel opens on the right, below the menus and tabs, so Undo, Search and the menus stay within reach. On a screen at least 1000 pixels wide, your data or output moves aside to make room for it, so nothing is hidden underneath. Drag the panel's left edge to make it wider. Close it with the cross at the top right or `Esc`. Your conversation stays while Socius is open, even when you switch tabs; reloading the page clears it.

If AI help is not set up yet and you send a question, the assistant does not lose it: the question stays in the box, and a note offers **Set up AI**. Once AI is set up, click **Send**.

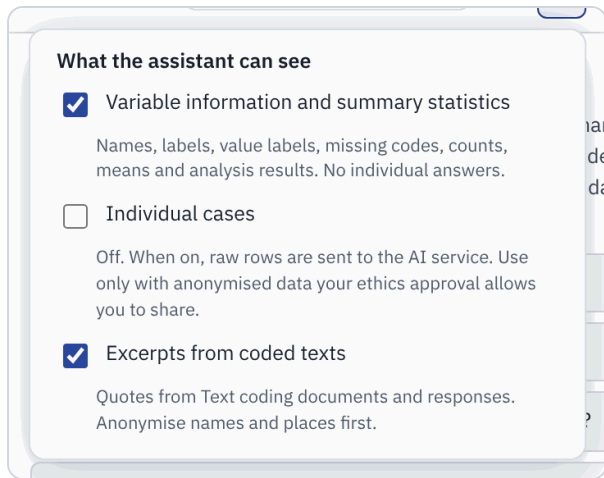
The screenshot shows the Socius software interface. The main window displays a dataset titled "Urban trust survey (sample)" with 640 cases and 34 variables. The "Data View" tab is active, showing a table of data. On the right, the "Assistant" panel is open, showing a conversation with Google Gemini (gemini-3.6-fla...). The panel includes a text input box with suggested questions, a "Send" button, and a status message at the bottom. Numbered callouts 1 through 4 highlight key elements: 1 points to the "Assistant" tab, 2 points to the eye icon, 3 points to the suggested questions, and 4 points to the "Send" button.

The assistant panel: the AI service in use (1), what the assistant can see (2), suggested questions (3) and the message box (4).

When the conversation is empty, the assistant suggests questions (3) that fit your data and the tab you are on. Click one to send it, or type your own question in the box (4) and press `Enter`. `Shift+Enter` starts a new line.

What it can see

Click the eye button (2) to see and change what the assistant may send to the AI service.



What the assistant can see. Individual cases are off unless you switch them on.

- **Variable information and summary statistics** (on): names, labels, value labels, missing codes, counts, means and analysis results. No individual answers.
- **Individual cases** (off): the raw rows of your data, up to 30 at a time. Switch this on only for anonymised data that your ethics approval lets you share. It switches off again every time you reload the page.
- **Excerpts from coded texts** (on): quotes from your Text coding documents and answers. Anonymise names and places first.

The line at the bottom of the panel always says which AI service answers, and that only what is listed here is sent.

Good first questions

With the sample survey open, try:

- "Describe my dataset"
- "Which test should I use to compare life satisfaction between migrants and non-migrants?"
- "Is trust in neighbours related to gender?"
- "Help me build a trust scale"
- "How do I export my results to Word?"

Ask one thing at a time and use your own words; you do not need variable names. You can ask follow-up questions ("Why not ANOVA?", "How do I report that?").

What it does with your question

The assistant does not guess. It checks your variables, runs the analysis in Socius on your data and reads the real tables before it answers.

Assistant Google Gemini (gemini-3.6-fla...)

Which test should I use to compare life satisfaction between migrants and non-migrants?

What I did (4 steps) 1

1. Looked at the dataset overview (34 variables, 640 cases); 1 possible issue
2. Let me check both variables and run the test.
3. Looked at life_sat, migrant
4. Ran Independent-Samples T Test: life_sat by migrant

Short answer

Use an **independent-samples t-test**. `life_sat` is a 0 to 10 score and `migrant` has two groups, so you are comparing two means.

What your data show

Group	N	Mean	SD
Born in this city	366	6.37	1.80
Migrated	264	5.70	1.85

People born in the city are somewhat more satisfied with life. The difference is statistically significant and small to medium: $t(628) = 4.51, p < .001, d = 0.36$. Levene's test ($p = .595$) is not significant, so read the row *Equal variances assumed*.

Next steps

1. Click **Add this analysis to Output** to keep the tables and the APA sentence.
2. Check these numbers against the tables in Output.
3. To take age or education into account as well, ask me about a linear regression.

Independent-Samples T Test
N = 630; 10 cases excluded for missing values.

Add this analysis to Output 2

Copy answer Retry

The assistant answers a question about life satisfaction. What I did (1) lists each step; the card (2) adds the t-test to Output. Example answer; your AI's wording will differ.

- **What I did** (1) lists every step it took, such as "Looked at the dataset overview", "Looked at life_sat, migrant" and "Ran Independent-Samples T Test: life_sat by migrant". Click it to see the steps.
- The answer quotes the real numbers, here $t(628) = 4.51, p < .001, d = 0.36$, the same as in Comparing two groups (page 22).
- An analysis the assistant runs is **not** added to Output by itself. Click **Add this analysis to Output** (2) to keep it.
- Some answers end with **Open dialog**, which opens the analysis dialog with the variables filled in. Check them and click **Run**.

When a question needs new variables, the assistant prepares the change and shows it to you first.

Assistant
Google Gemini (gemini-3.6-fla...)
Help me build a trust scale

What I did (4 steps)

1. Looked at the dataset overview (34 variables, 640 cases); 1 possible issue
2. Let me look at the five trust items first.
3. Looked at trust1, trust2, trust3, trust4, trust5
4. Proposed: Reversed trust3 to trust3_r.

First, reverse one item

trust3 ("You have to be very careful with people in this neighbourhood") is worded the other way round: agreeing means **less** trust. If you add it as it is, it pulls the scale apart and lowers Cronbach's alpha.

I prepared **trust3_r** below: 1 becomes 5, 2 becomes 4, and so on. The missing codes 8 and 9 stay missing.

Next steps

1. Check the preview and click **Apply**.
2. Then ask me to build the scale from trust1, trust2, trust3_r, trust4 and trust5. I will check its alpha.

1
Proposed change: Reverse-code Items
Reversed trust3 to trust3_r. Nothing changes until you click Apply.

Case	trust3	trust3_r (new)
1	4 (Agree)	2 (Agree)
2	5 (Strongly agree)	1 (Strongly agree)
3	3 (Neither agree nor disagree)	3 (Neither agree nor disagree)
4	2 (Disagree)	4 (Disagree)
5	4 (Agree)	2 (Agree)
6	2 (Disagree)	4 (Disagree)

► SPSS syntax

2
Apply
Dismiss

Copy answer
Retry

Asked to build a trust scale, the assistant first proposes to reverse trust3 (1). Nothing changes until you click Apply (2). Example answer; your AI's wording will differ.

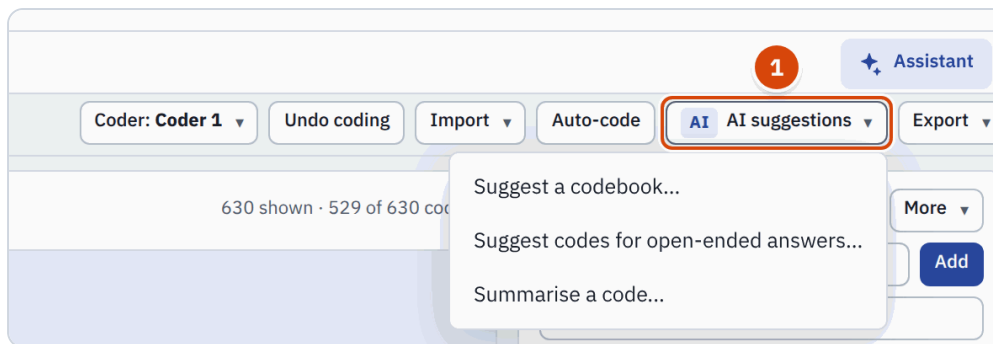
A **Proposed change** card (1) describes the recode, reverse-coding, scale or computed variable, shows the first cases before and after, and gives the SPSS syntax. Nothing changes until you click **Apply** (2); **Dismiss** throws the proposal away. After Apply, the change is logged in Output like any Transform command, and **Edit** > **Undo** reverses it. The assistant only creates new variables; it never overwrites your data.

Know its limits

- **It can be wrong.** It may misread a table, pick an unsuitable test or phrase a finding too strongly. Check its numbers against the tables in Output, and think about whether its advice fits your research question.
- **Free tiers have limits.** Google's free tier allows only a certain number of requests a minute and a daily allowance. The assistant paces itself, so a complex question can take 20 to 60 seconds. When Google asks it to slow down, the line under your question counts down, for example "Waiting 11 s for Google's free limit...", and then carries on by itself. If you see "Too many AI requests at once, or the free allowance is used up for now", wait a minute and click **Retry**. When the daily allowance is used up, the message says so; it starts again at midnight Pacific time (morning in Europe, early afternoon in India).
- **The on-device model is weaker.** On this computer, the assistant works with a small model: slower, with fewer tools and simpler answers. For the assistant, Gemini works much better. Use anonymised data with it.
- **It cannot replace your judgement** or your supervisor. It is a tutor at your side, not an authority.

AI in Text coding

The coding helpers live in the **AI** menu. In the Text coding tab, the **AI suggestions** button in the toolbar is a shortcut to the same three items. If AI is not set up, the button is greyed out and a **Set up AI** link sits next to it.



The AI suggestions menu (1) in the Text coding toolbar.

- **Suggest a codebook** reads a sample of your documents or answers and proposes codes, each with a definition and an example quote. Keep the ones you want and click **Add** (the button shows how many codes). Treat them as a first draft: rename, merge and define them in your own terms.
- **Suggest codes for open-ended answers** applies your existing codebook to open-ended answers and suggests codes for each one. You **Accept** or **Reject** every suggestion, or **Accept all**. Accepted codes are marked as AI suggestions, so you can review them later.
- **Summarise a code** opens the **Retrieve** view. Pick a code and click **Summarise this code** for a short summary of its passages. **Save as memo** keeps it as a memo on the code. Check it against the quotes.

Each dialog says how many excerpts will be sent, and to whom, before you click. For intercoder reliability, AI suggestions do not count as a second coder: a person must code independently.

Reporting AI use

If AI helped with your analysis, say so in your methods section, as you would for any software. Name the tool and the model, say what it did and how you checked it. For example:

AN EXAMPLE FOR YOUR METHODS SECTION

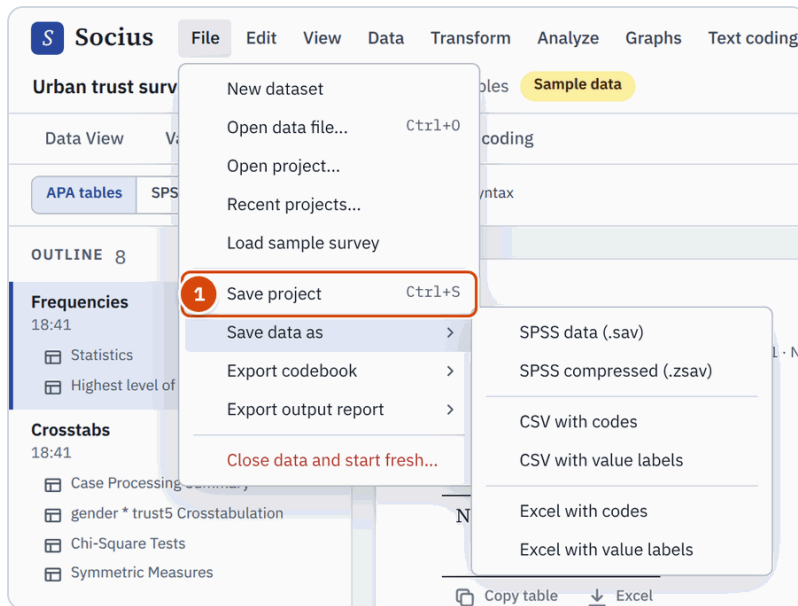
Open-ended answers were coded in Socius. An initial codebook was drafted with AI assistance (Google Gemini, via Socius's "Suggest a codebook"), then revised by the author; all codes were applied and checked by hand. Plain-language explanations of statistical output were generated with the same model and checked against the output tables. No identifiable data were sent to the AI service.

Follow your university's and your journal's rules on AI, which differ and change often. Keep a note of what you used AI for while you work; it is much harder to remember afterwards.

Saving and sharing your work

Save a project

A project file keeps everything together: the data (including the variables you created), all your output and the text coding. Choose **File > Save project** (or press `Ctrl+S`). Socius saves one file ending in `.socius.json` to your downloads folder.



The File menu: Save project (1) and Save data as, with its formats.

To continue later, on this or any other computer, choose **File > Open project** and pick the file. **File > Recent projects** lists projects you saved or opened in this browser.

AUTOSAVE IS A CONVENIENCE, NOT A BACKUP

Socius keeps a copy of your current session in the browser and restores it when you come back. But that copy disappears if you clear your browser's history or site data, use a private or incognito window, or switch to another browser or computer. Save a project file regularly and keep it with your other research files.

If the browser runs out of room (usually after downloading an on-device AI model), autosave pauses and a bar at the bottom says **Browser storage is full, so autosave is paused**. Click **Save project** to save your work to a file straight away, then **Manage storage** to delete a downloaded AI model you do not need. Autosave keeps trying by itself and the bar disappears as soon as a save works again.

What Socius keeps in your browser

Help > About Socius lists what Socius stores in this browser, under "What Socius stores in this browser":

- **Autosave:** the data, output and text-coding project you have open, and your list of recent projects.

- **Preferences:** the theme, view settings and your search history.
- **Error log:** technical messages only, without data values, names or keys (see Troubleshooting with the error log (page 61)).
- **AI assistant settings:** the AI option and model you chose, and your API key only if you ticked **Remember this key on this computer**.

The About box also points out a limit of the free website: browsers treat all websites published on the same github.io account as one site, so another website published there could read what Socius stores. To keep confidential work safe, save it to project files instead of relying on autosave, use **File > Close data and start fresh** when you finish on a shared or public computer, and leave the remember option for your key off.

Under **Storage used**, the About box shows how much room Socius takes and how much the browser still allows, split into downloaded AI models (each with a **Delete** button), projects and autosave, and the error log and settings.

WHAT SOCIUS STORES IN THIS BROWSER

- **Autosave:** the data, output and text-coding project you have open, and your recent projects, so you can pick up where you left off.
- **Preferences:** theme, view settings and the search history.
- **Error log:** technical messages only, without data values, names or keys (Help > Error log).
- **AI assistant settings:** the provider and model you chose, and your API key only if you ticked the remember option.

Socius is published at **hackhead95.github.io**. Browsers treat every website published on that github.io account as one site, so another website hosted there could read what Socius stores in this browser.

To keep confidential work safe:

- Save confidential projects to files (File > Save project) instead of relying on the browser's autosave.
- On a shared or public computer, use **File > Close data and start fresh** when you finish.
- Leave the remember option for your API key off. It is in **AI > AI assistant settings**, the one place where AI is set up.

STORAGE USED

This site uses **12 MB** of the **1.0 GB** the browser allows here; **1.0 GB free**.

AI model: Small and fast Qwen2.5-1.5B-Instruct-q4f16_1-MLC	12 MB	Delete
Projects and autosave	80 KB	
Error log and settings	5 KB	

Delete downloaded AI models
Check again
Kept when disk space runs low: no, the browser may clear it. [Ask the browser to keep it](#)

Help > About Socius: what Socius stores in this browser, and how much room it uses.

Ask the browser to keep it asks the browser not to clear Socius's data when the disk runs low. Browsers decide this themselves, so it may say no.

Save your data for SPSS, Excel or R

File > Save data as writes your data in other formats:

- **SPSS data (.sav):** opens in SPSS with all labels, value labels and missing values, including the variables you created in Socius. **SPSS compressed (.zsav)** is a smaller file for SPSS 21 or later.
- **CSV with codes** or **CSV with value labels:** plain text that almost any program reads, including R and Stata.
- **Excel with codes** or **Excel with value labels:** a spreadsheet.

File > Export codebook saves a list of all variables with their labels, value labels and missing values, handy for a thesis appendix.

Share with a colleague

Send your colleague the `.socius.json` project file (by email or a shared drive). They open Socius and choose **File > Open project**. If they use SPSS, send the `.sav` file instead. Under every result in Output, the **Syntax** section holds the SPSS commands that reproduce it.

CHAPTER 14

Troubleshooting and FAQ

My file will not open. Socius opens `.sav`, `.zsav`, `.csv`, `.tsv`, `.txt` and `.xlsx` files. For other formats:

- Old Excel files (`.xls`): open in Excel and save as `.xlsx` or CSV.
- SPSS portable files (`.por`): open in SPSS or PSPP and save as `.sav`.
- Stata, SAS or R files: export them as CSV, or save as `.sav` (in R, `haven::write_sav()`).
- SPSS output (`.spv`) and syntax (`.sps`) files are not data. Open the `.sav` file instead.
- Zipped files: unzip first.

Letters look wrong (strange symbols instead of Hindi, Bengali or accented letters). This is a text encoding problem. For a CSV file, choose another **Character encoding** in the preview. In Excel, saving as "CSV UTF-8" avoids the problem. Recent SPSS files open correctly; for a very old one, open and save it again in a recent SPSS.

My results differ from SPSS. Check that the same codes are declared missing (Variable View, Missing column), and that the same weight and filter are on. The line under each result title says how many cases were used and whether they were weighted.

A dialog warns that a variable has the wrong measurement level. It is a warning, not a block. Often the fix is to set the right level in the Measure column of Variable View.

I made a mistake. **Edit > Undo** (`Ctrl+Z`) reverses your last change in the tab you are in: changes to the data in Data View and Variable View, coding changes in Text coding, and a deleted result in Output. **Edit > Redo** (`Ctrl+Y`) brings it back. The Undo item says what it will undo, for example "Undo rename of age" or "Undo new variable agegrp". While you type in a box, `Ctrl+Z` undoes your typing. Text coding also has an **Undo coding** button in its toolbar.

I lost my work after clearing the browser. Clearing browsing data, a private window, or another browser or computer loses the autosaved session. Only a saved project file is safe. Save one regularly with **File > Save project**.

Where is my data stored? Only in your browser, on your computer. Files you open are never uploaded. The only things that can leave your computer are what you choose to send with the optional online AI help (see Before you send anything (page 48)). **Help > About Socius** lists what Socius keeps in the browser (see What Socius keeps in your browser (page 57)).

A bar says "Browser storage is full, so autosave is paused". The browser has no room left for Socius, usually because of a downloaded on-device AI model. Click **Save project** first, then **Manage storage** and delete the AI model you do not use. Autosave starts again by itself. See Save a project (page 57).

A message says "Socius was updated". A new version of Socius was published while you were working. Click **Reload**: your work is autosaved first, and the page opens again with the new version.

I set up AI but I don't see a way to use it. Use the **AI** menu in the menu bar, the **Assistant** button at the right end of the tabs (or `Ctrl+J`), **Explain with AI** on any result in Output, or **AI suggestions** in the Text coding toolbar. The **AI** chip at the top right lists every AI feature. See [Where to find AI in Socius \(page 48\)](#).

Test connection fails. Look at the list of steps: the first step with a cross shows where it stopped, and the red box below says what to do (see [Test connection, step by step](#)).

- **1. Internet connection** failed: this computer is offline. Connect, then test again.
- **2. Reached Google** failed ("Could not reach the AI service"): something between you and Google blocks it. Common causes are an ad or privacy blocker, antivirus web protection, a university or company firewall, or a VPN. Turn these off for this site, or try another network.
- **3. Key accepted** failed ("The AI service did not accept the key"): copy the key again from Google AI Studio with its copy button, and paste it into **API key**. A brand-new key can take a few minutes to work. If it still fails, create a new key.
- **4. Model chosen** failed: set **Model** to **Automatic: Flash-Lite**, so that Socius picks a model your key may use.
- Google says your location is not supported: the free Gemini API is not offered everywhere. If you use a VPN, turn it off; otherwise use **On this computer**.
- Google AI Studio asks you to be 18 or older. If you cannot create a key, use **On this computer** instead.

If you cannot solve it, click **Copy details** and send the report with **Help > Send feedback or report a problem**. The report never contains your key.

Ollama or LM Studio does not connect. Use **Test connection** in the Ollama or LM Studio set-up: it checks each step and shows the fix. The most common one is "Ollama is running but refused this website": set [OLLAMA_ORIGINS](#) as the app shows and restart Ollama (see [Option 3](#)). In Safari, use Chrome, Edge or Firefox instead.

AI replies are slow. Check that **Model** is **Automatic: Flash-Lite**, the fastest choice. The line under a request shows what it is waiting for. "Waiting ... for Google's free limit" means the free tier's per-minute limit was reached; Socius waits and continues by itself. The assistant often needs several steps for one question, so 20 to 60 seconds is normal. The on-device model is much slower than Gemini, especially the first time, while it loads.

The AI says the free allowance is used up ("rate limit"). Free tiers allow only a certain number of requests a minute and a limited number a day. For the per-minute limit, wait a minute and click **Retry** (or **Try again**). If the message says the daily allowance is used up, it starts again at midnight Pacific time (morning in Europe, early afternoon in India). Meanwhile, send fewer excerpts at a time, or switch to **On this computer**.

"This browser cannot run the on-device model" (no WebGPU). The on-device model needs WebGPU, which only recent Chrome and Edge on a desktop or laptop offer reliably. The settings explain what is missing on your computer. Update your browser, or use another computer. Check that graphics acceleration is on (Chrome: Settings > System > "Use graphics acceleration when available"). On Linux, Ollama ([Option 3](#)) is usually the more reliable private choice. Otherwise use Google Gemini with anonymised data.

The model download fails or says there is not enough space. A failed download can be tried again; the parts that already arrived are kept. If the browser has no room, delete a model you no longer use (**Browser storage (downloaded models)** in the settings, or **Help > About Socius**), choose **Small and fast**, or free disk space. Private windows allow very little storage, so use a normal window.

AI buttons are greyed out. In Text coding, a grey **AI suggestions** button means AI help is not set up: click **Set up AI** next to it. If AI is set up but a single item is grey, it has nothing to work on yet: **Suggest a codebook** needs imported texts, **Suggest codes for open-ended answers** needs imported answers and at least one code, and **Summarise a code** needs coded passages. The **Explain** button in the Explain panel stays grey until AI help is set up.

The assistant's numbers differ from Output. Trust the tables in Output. The assistant runs the same analyses, but it can misread or round them. Run the analysis from the menu (or click **Add this analysis to Output**) and report the numbers from Output.

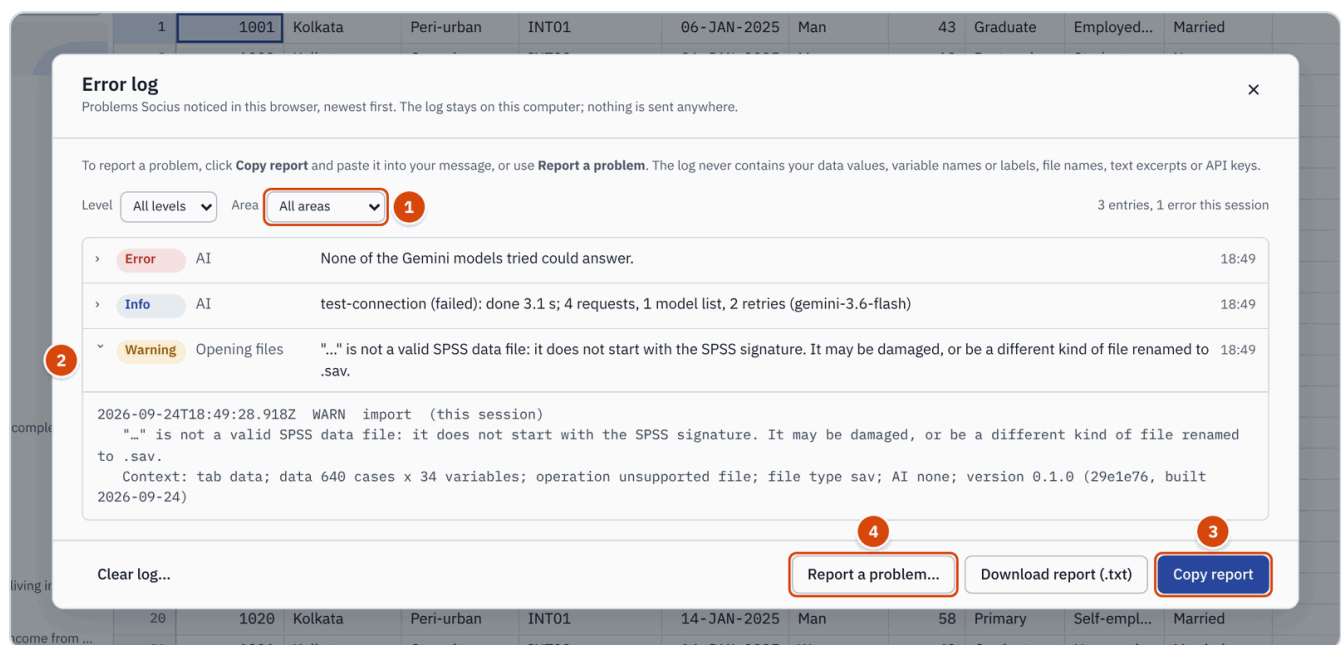
Can I use Socius on a phone? Yes, for looking at data and results. For coding and analysis, a computer with a mouse and keyboard is much easier. On a small screen the menus are in a **Menu** button, the variable list opens from **View > Variable list** as a drawer, and the assistant fills the screen.

I clicked outside a dialog and it did not close. Dialogs with boxes to fill in stay open, so a stray click does not lose your choices. Click **Cancel** or press Esc.

Where are the keyboard shortcuts? **Help > Keyboard shortcuts** lists them all, including the keys of the Responses view in Text coding. The two most useful are Ctrl+K for Search and Ctrl+J for the assistant. **Help > Getting started** gives a six-step overview.

Troubleshooting with the error log

When something goes wrong, Socius writes a short note in its **error log**, so you can tell us exactly what happened. Open it with **Help > Error log**. After a new error, a small blue dot appears next to **Help** in the menu bar; opening the log removes it.



The error log. Filter by level or area (1), click a line to see its details (2), then Copy report (3) or Report a problem (4).

- Each line shows a level (**Error**, **Warning** or **Info**), the area (such as AI, Opening files or Analyses), the message and the time. Click a line (2) to see the technical details.
- **Level** and **Area** (1) filter the list.
- **Copy report** (3) copies the whole log as text; **Download report (.txt)** saves it as a file.
- **Report a problem...** (4) opens the feedback dialog (see Giving feedback (page 62)).
- **Clear log...** deletes all entries, after asking.

The log stays in your browser; nothing is sent anywhere unless you paste it into a message yourself. It never contains your data values, variable names or labels, file names, quotes from your texts, email addresses or API keys: Socius removes them before anything is written. It keeps only the message, the kind of error, the size of the dataset (such as "640 cases x 34 variables"), which tab was open, the AI service and model, and the Socius version.

IF SOCIUS STOPS WORKING

If a whole tab stops working, it shows a message with **Try again**, **Copy error report** and **Open the error log**; your data and the other tabs are not affected. If the whole page fails, a card says **Something went wrong**. Click **Copy error report** first, then **Reload**: your work is autosaved in the browser, although the last few seconds may be missing.

CHAPTER 15

Giving feedback

Socius is young, and your experience helps make it better. If something does not work, confuses you, or you miss a feature, please tell us.

Choose **Help > Send feedback or report a problem**, or click **Feedback** at the top right of the screen. A small dialog opens first:

1. **Copy the error report.** It says how many problems are in the error log. Click **Copy error report**, and paste the report into your message later; it helps us find the cause quickly.
2. **Open the feedback form.** Keep **Fill in a short summary: version, browser and the last 5 problems** ticked to add that summary to the form for you (you can read it in the dialog first). Click **Open the feedback form** to report a problem, or **Suggest an idea instead**. The form opens on GitHub in a new tab.

Send feedback or report a problem

×

The form opens on GitHub in a new tab (a free GitHub account is needed).

1. COPY THE ERROR REPORT

The error log has 2 problems (2 from this session). Copy the error report to paste into your message: it helps us find the cause quickly.

Copy error report

Open the error log

2. OPEN THE FEEDBACK FORM

☒ Fill in a short summary: version, browser and the last 5 problems

Version: 0.1.0 (29e1e76, built 2026-09-24)

Browser: Chrome 141 on Linux

Page: <https://hackhead95.github.io/socius/>

Recent problems (newest first, 2 of 2):

- 2026-09-24 18:49 error ai: None of the Gemini models tried could answer. [unavailable, test-connection]

- 2026-09-24 18:49 warn import: "... is not a valid SPSS data file: it does not start with the SPSS signature. It may be damaged, or be a different kind of file renamed to .sav. [unsupported file]

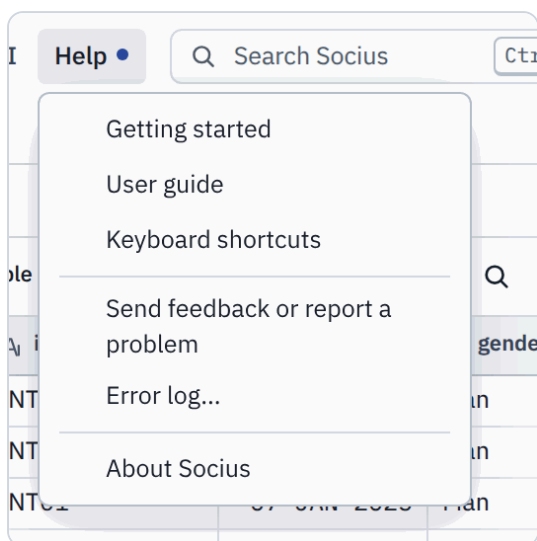
The report and summary never include your data values, variable names or labels, file names, text excerpts or keys. Please do not paste confidential or identifiable research data into the form.

Suggest an idea instead

Open the feedback form

The feedback dialog: copy the error report, then open the form on GitHub.

You can also go to the form directly: <https://github.com/hackhead95/socius/issues/new/choose>



The Help menu, with the user guide, the keyboard shortcuts, the feedback dialog and the error log. The blue dot next to Help means a new error was logged. AI set-up is in the AI menu.

Help > About Socius shows the version you are using, for example "Version 0.1.0 (29e1e76, built 2026-09-24)". The error report includes it, so you do not need to copy it yourself.

- Posting on GitHub needs a free GitHub account. Creating one takes a minute.
- Say what you did, what you expected and what happened instead. A screenshot helps, and so does the error report.
- Never attach confidential data. If a problem only happens with your file, describe the file (for example "an SPSS file with 300 variables from 2012") instead of sending it.

Thank you for trying Socius.

Quick reference card

I want to...	Where to find it
Find any command, variable or result	Search box at the top, or Ctrl+K
Go back to the start screen	The Socius logo at the top left
Open an SPSS, CSV or Excel file	File > Open data file
See labels, missing values, measure	Variable View tab
Rename a variable	Variable View , Name column (or double-click the column heading in Data View)
Label values and find missing codes in many variables	Data > Define variable properties
Show labels instead of codes in the data	View > Value labels in Data View
Copy labels to other variables	Data > Copy variable properties
Group ages or incomes	Transform > Recode into different variables
Flip a negatively worded item	Transform > Reverse-code items
Build a scale with Cronbach's alpha	Transform > Create scale / index
Compute a new variable with a formula	Transform > Compute variable
Analyse only some cases	Data > Select cases
Use a survey weight	Data > Weight cases
Count answers	Analyze > Descriptive Statistics > Frequencies
Mean, SD, minimum, maximum	Analyze > Descriptive Statistics > Descriptives
Relate two categorical variables	Analyze > Descriptive Statistics > Crosstabs
Compare two group means	Analyze > Compare Means > Independent-Samples T Test
Compare three or more group means	Analyze > Compare Means > One-Way ANOVA
Correlate scale variables	Analyze > Correlate > Bivariate Correlations
Predict a scale outcome	Analyze > Regression > Linear Regression
Predict a yes/no outcome	Analyze > Regression > Binary Logistic Regression
Check a scale's reliability	Analyze > Scale > Reliability Analysis
Make a bar chart or histogram	Graphs > Bar Chart, Graphs > Histogram
Copy a table into Word	Copy table under the table in Output
Export all results to Word	File > Export output report > Word document (.docx)
Code open-ended answers	Text coding > Import open-ended answers from dataset
Import interview transcripts	Text coding > Import documents
Auto-code with keywords	Text coding > Auto-code with keyword rules
Turn codes into variables	Text coding > Export codes to dataset
Check agreement between coders	Text coding > Intercoder reliability
Set up AI	AI > AI assistant settings
Ask a question about my data or methods	Assistant button (right end of the tabs), or Ctrl+J
Explain a result in plain language	Explain with AI on the result in Output

I want to...	Where to find it
Save everything	File > Save project (Ctrl+S)
Save data for SPSS	File > Save data as > SPSS data (.sav)
Undo (in the tab you are in)	Edit > Undo (Ctrl+Z)
Report a problem	Help > Send feedback or report a problem
See what went wrong	Help > Error log
See what Socius stores in the browser	Help > About Socius

Glossary

Term	Meaning
Case	One row of data, usually one respondent.
Variable	One column of data, usually one question.
Value label	The words behind a code, such as 2 = Woman.
Missing value	A code that means "no real answer", such as 9 = Refused. Left out of statistics.
Nominal, ordinal, scale	Measurement levels: unordered categories; ordered categories; real numbers.
p-value	The probability of a result at least this strong if there were really no relationship. Below .05 is usually called statistically significant. It says nothing about how big or important an effect is.
Statistically significant	Unlikely to be due to chance alone (usually $p < .05$).
Chi-square (χ^2)	A test of whether two categorical variables are related, used with crosstabs.
Cramér's V	The strength of a relationship in a crosstab, from 0 (none) to 1 (perfect). About .10 is weak, .30 moderate, .50 strong.
Mean (M) and standard deviation (SD)	The average, and how far values typically lie from it.
t-test	A test of whether the means of two groups differ.
ANOVA	Analysis of variance: a test of whether the means of three or more groups differ.
Effect size	How big a difference or relationship is, regardless of sample size. Examples: Cohen's d, eta squared (η^2), Cramér's V, r.
Cohen's d	The difference between two means in standard deviation units. About 0.2 is small, 0.5 medium, 0.8 large.
Correlation (r)	How closely two variables move together, from -1 to +1. About .10 is weak, .30 moderate, .50 strong.
Regression coefficient (B)	How much the outcome changes for a one-unit increase in a predictor, holding other predictors constant.
R squared (R^2)	The share of the differences in an outcome that a regression model explains.
Cronbach's alpha	How consistently a set of items measures the same thing, from 0 to 1. .70 or higher is usually acceptable.
Degrees of freedom (df)	A number based on sample size and groups that is reported with a test statistic, as in t(628).
Code	A label for a theme that you attach to a passage of text.
Codebook	The list of your codes with a definition, inclusion and exclusion rules and an example for each.
Theme	A group of related codes (a parent code).
Memo	Your analytic note on a passage, a code or the project.
Intercoder reliability	How far two coders who code the same material independently agree, beyond chance. Measured with Cohen's kappa (κ) or Krippendorff's alpha (α); .70 to .80 or higher is usually expected.
APA style	The reporting style of the American Psychological Association (7th edition), used by many social science journals.

Term	Meaning
Syntax	The SPSS commands that reproduce a result. Socius shows them under each result.
AI model	A computer program trained on large amounts of text that writes answers in ordinary language. It can be wrong, so check what it says.
API key	A long password from an AI provider, such as Google, that lets Socius use its AI model on your behalf. Keep it private.
Socius assistant	The chat panel (Ctrl+J) where an AI model answers your questions, using Socius to look at your data and run analyses.
Search	The box at the top (Ctrl+K) that finds commands, variables, results and help pages.
Error log	Socius's list of problems it noticed in this browser (Help > Error log), without your data, to send with a problem report.
Rate limit	The number of requests a free AI service allows per minute or per day.
Figure	A numbered chart in APA style (Figure 1, Figure 2...), like a numbered table.

Socius beginner's guide. Edition 1, September 2026. The sample survey and interviews are fictional teaching material.